

Physical Chemistry (I)

Lecture 19

Special Lecture:

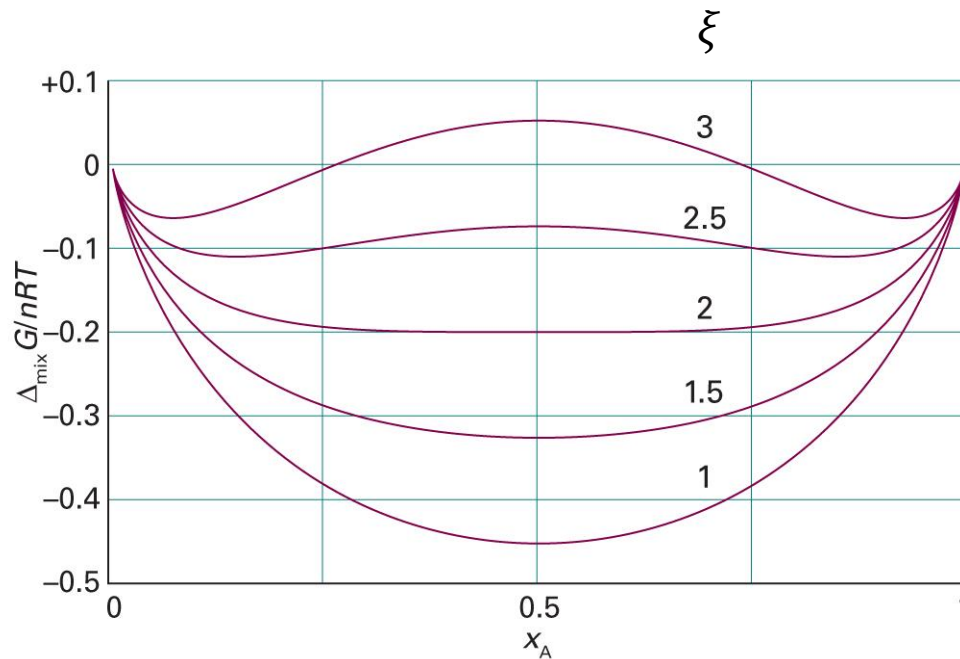
Phase Separation of Biomolecules



Last class

BW Model: Gibbs Energy of Mixing

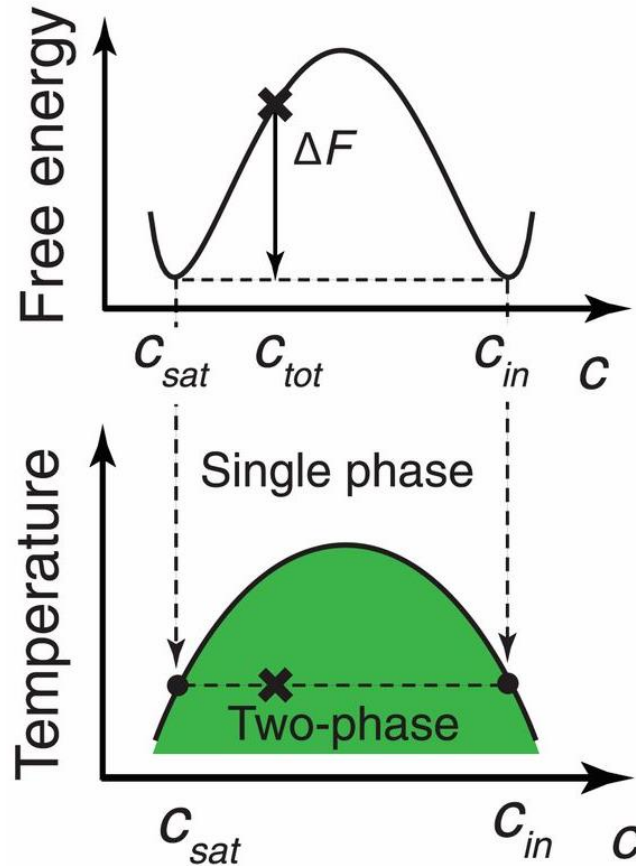
$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.5)

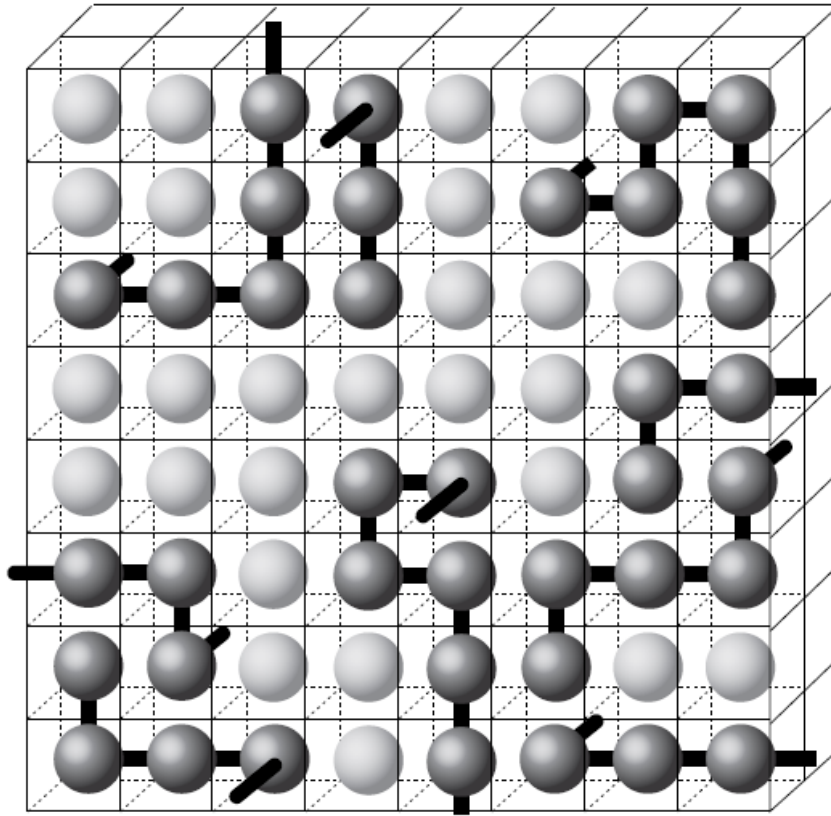
Last class

What Happens at $\xi > 2$?



(Image: Shin and Brangwynne, *Science* 357: eaaf4382 (2017), Figure 3)

Lattice Model of Polymers



M sites

N_s solvent molecules

N_p polymer molecules

L monomers / polymer

Entropy of Mixing

$$\Delta_{\text{mix}}S = -k_B \left(N_s \ln \frac{N_s}{M} + N_p \ln \frac{LN_p}{M} \right)$$

Define volume fractions:

$$\phi_s = \frac{N_s}{M}, \quad \phi_p = \frac{LN_p}{M}$$

Hence,

$$\frac{\Delta_{\text{mix}}S}{M} = -k_B \left(\phi_s \ln \phi_s + \frac{\phi_p}{L} \ln \phi_p \right)$$

Thermodynamics of Mixing

- Entropy of mixing

$$\frac{\Delta_{\text{mix}}S}{M} = -k_B \left(\phi_s \ln \phi_s + \frac{\phi_p}{L} \ln \phi_p \right)$$

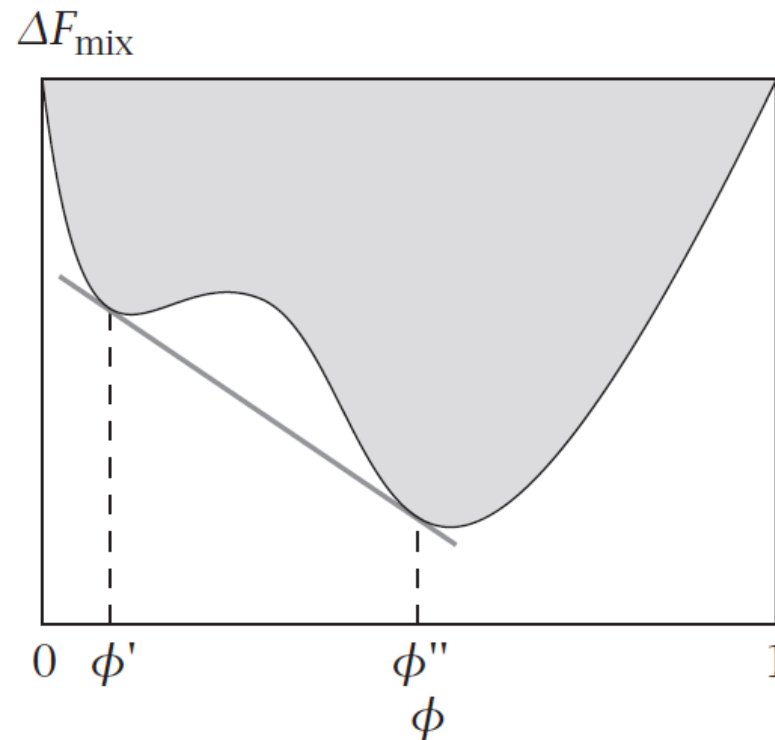
- Energy of mixing

$$\Delta_{\text{mix}}U = k_B T \xi_{\text{sp}} \frac{N_s N_p L}{M} = M k_B T \xi_{\text{sp}} \phi_s \phi_p$$

- Helmholtz energy of mixing

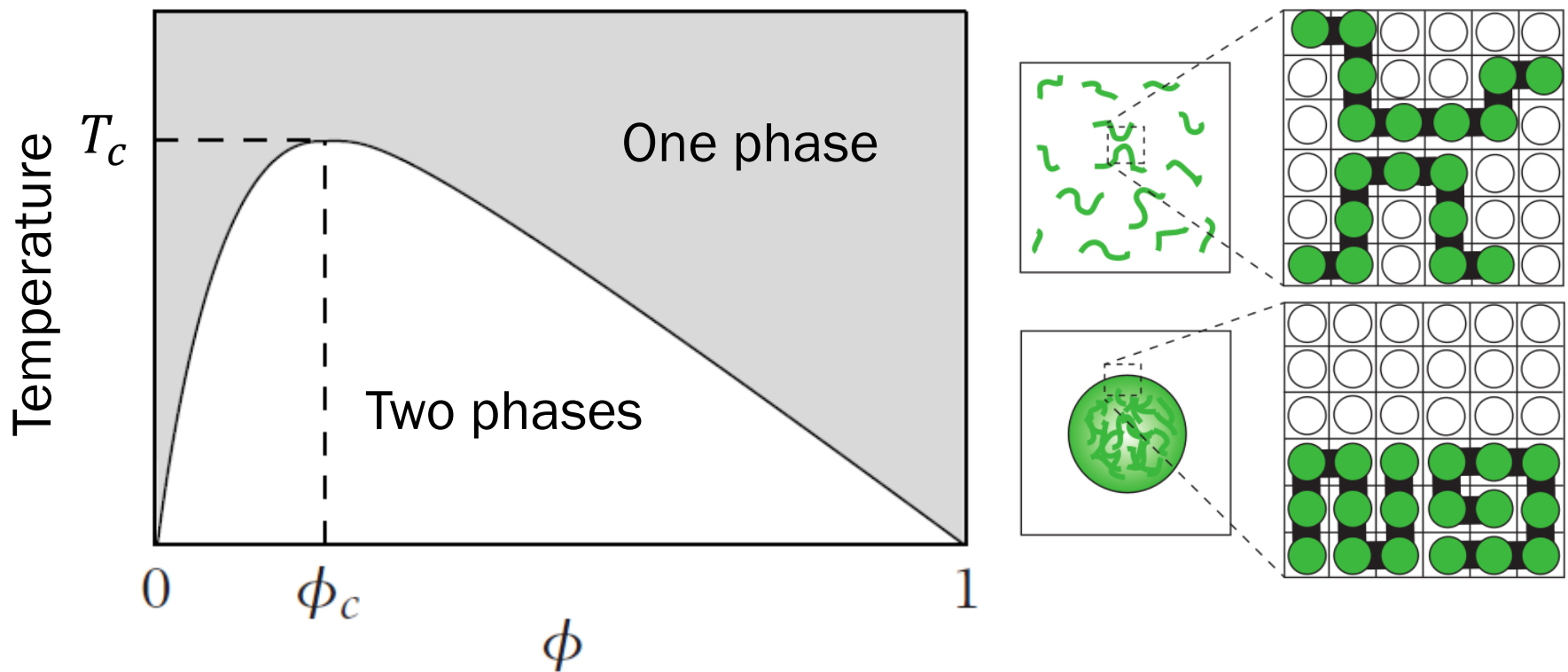
$$\frac{\Delta_{\text{mix}}F}{M k_B T} = \phi_s \ln \phi_s + \frac{\phi_p}{L} \ln \phi_p + \xi_{\text{sp}} \phi_s \phi_p$$

Free Energy of Mixing

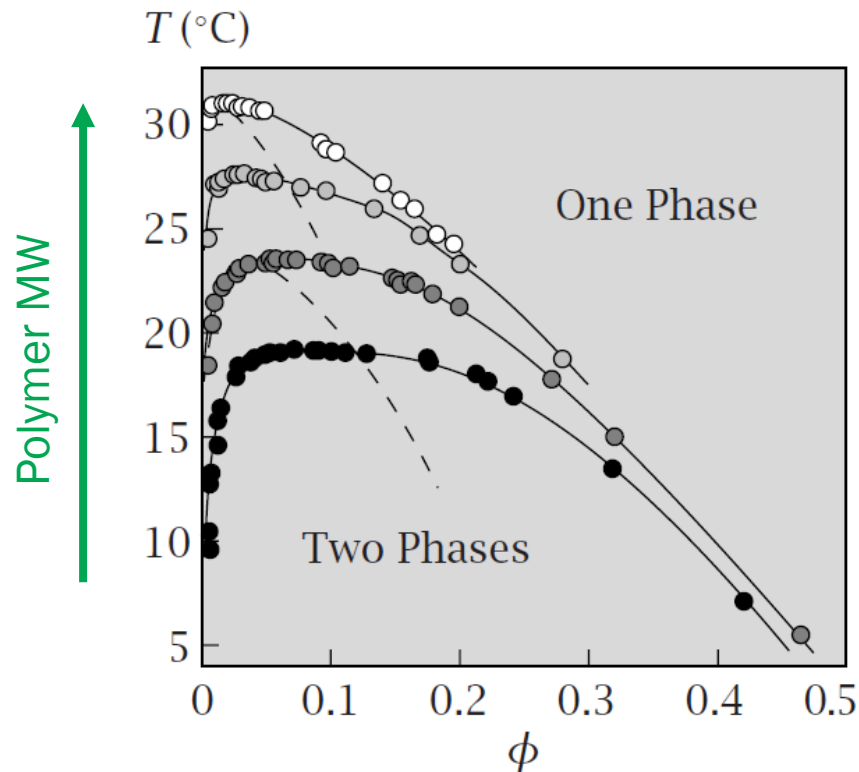


(Image: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figures 32.6)

Phase Diagram

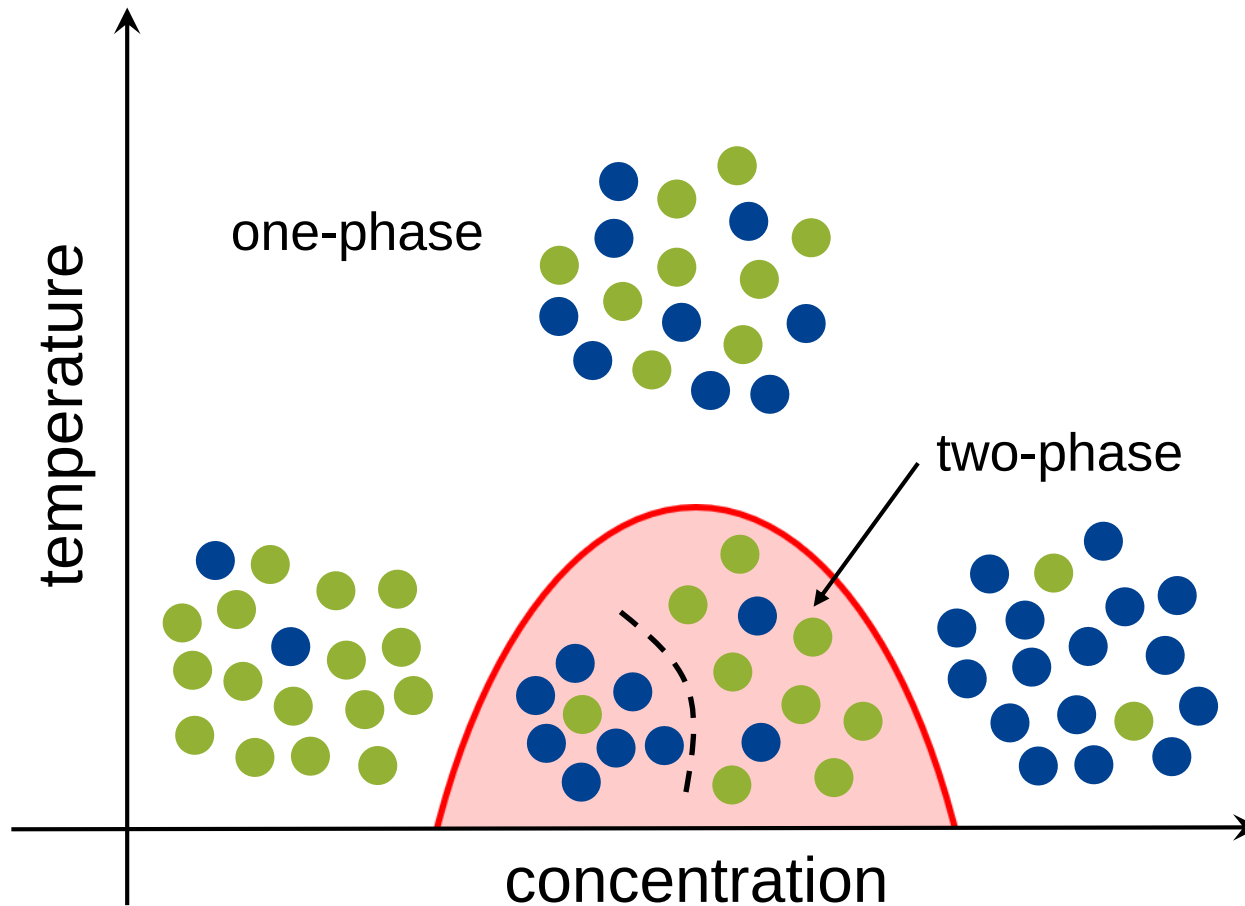


Polystyrene in Cyclohexane

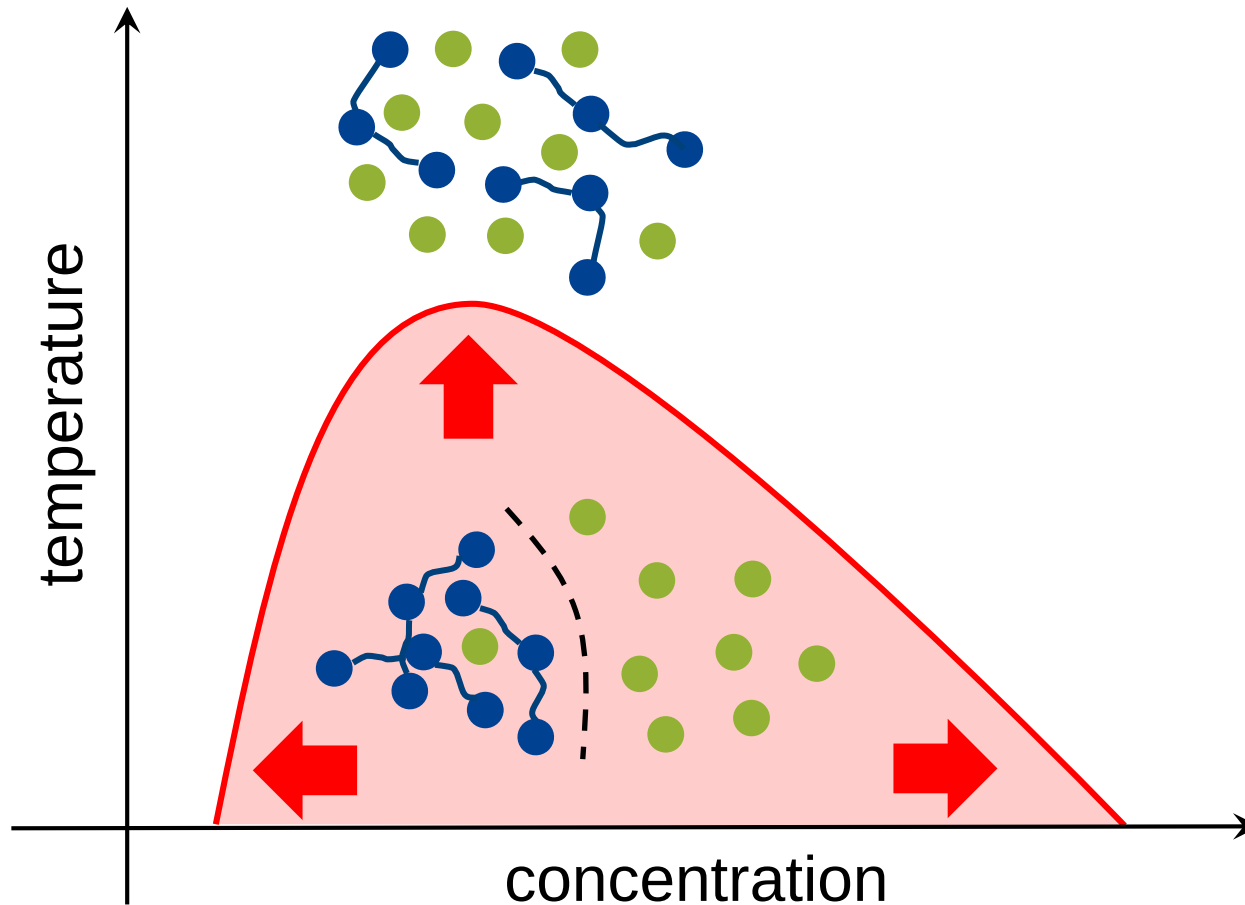


(Image: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figures 32.8)

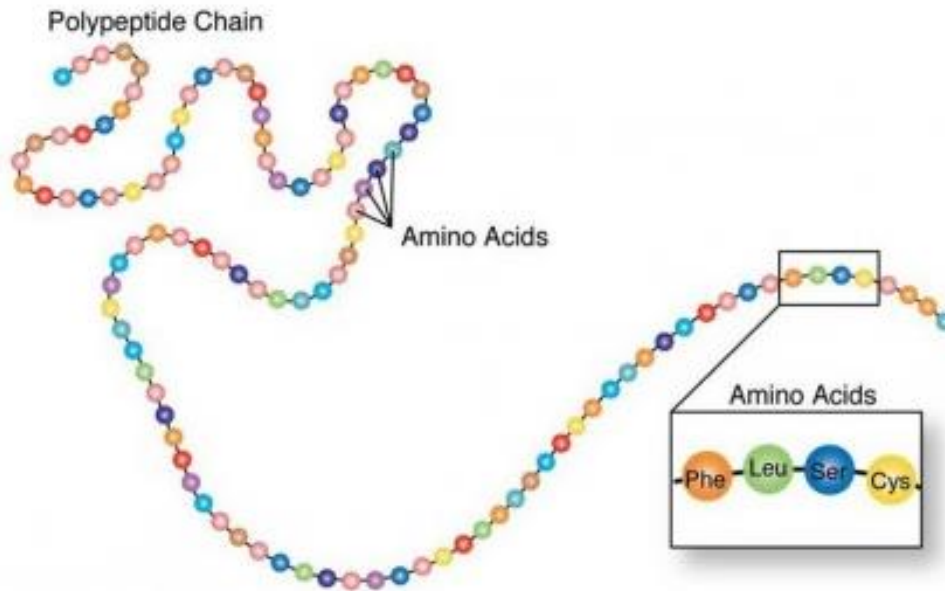
Phase Diagram for Monomers



In Case of Polymer + Solvent



Proteins are Polymer Chains



Proteins are Polymer Chains

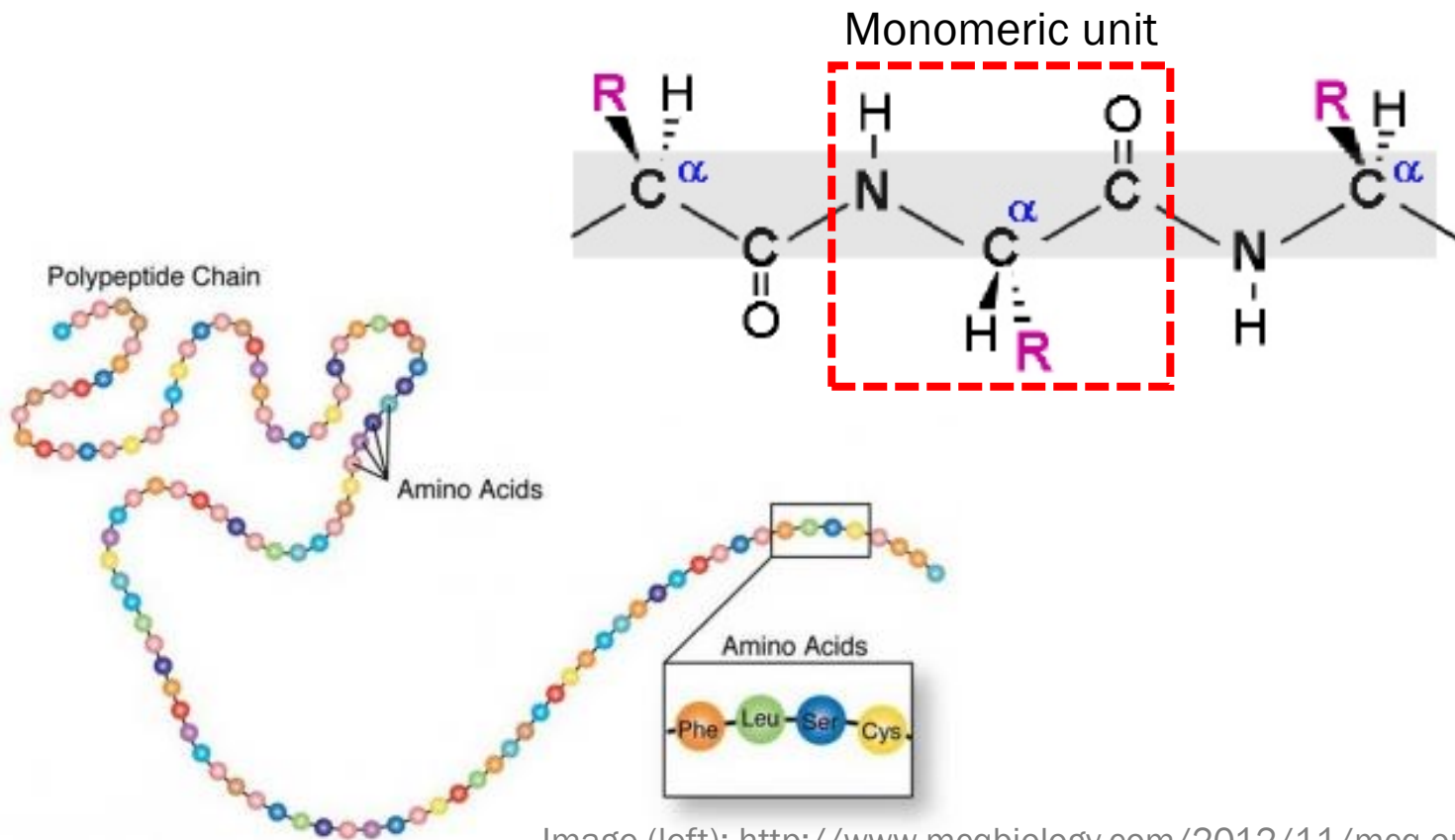


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>

Proteins are Polymer Chains

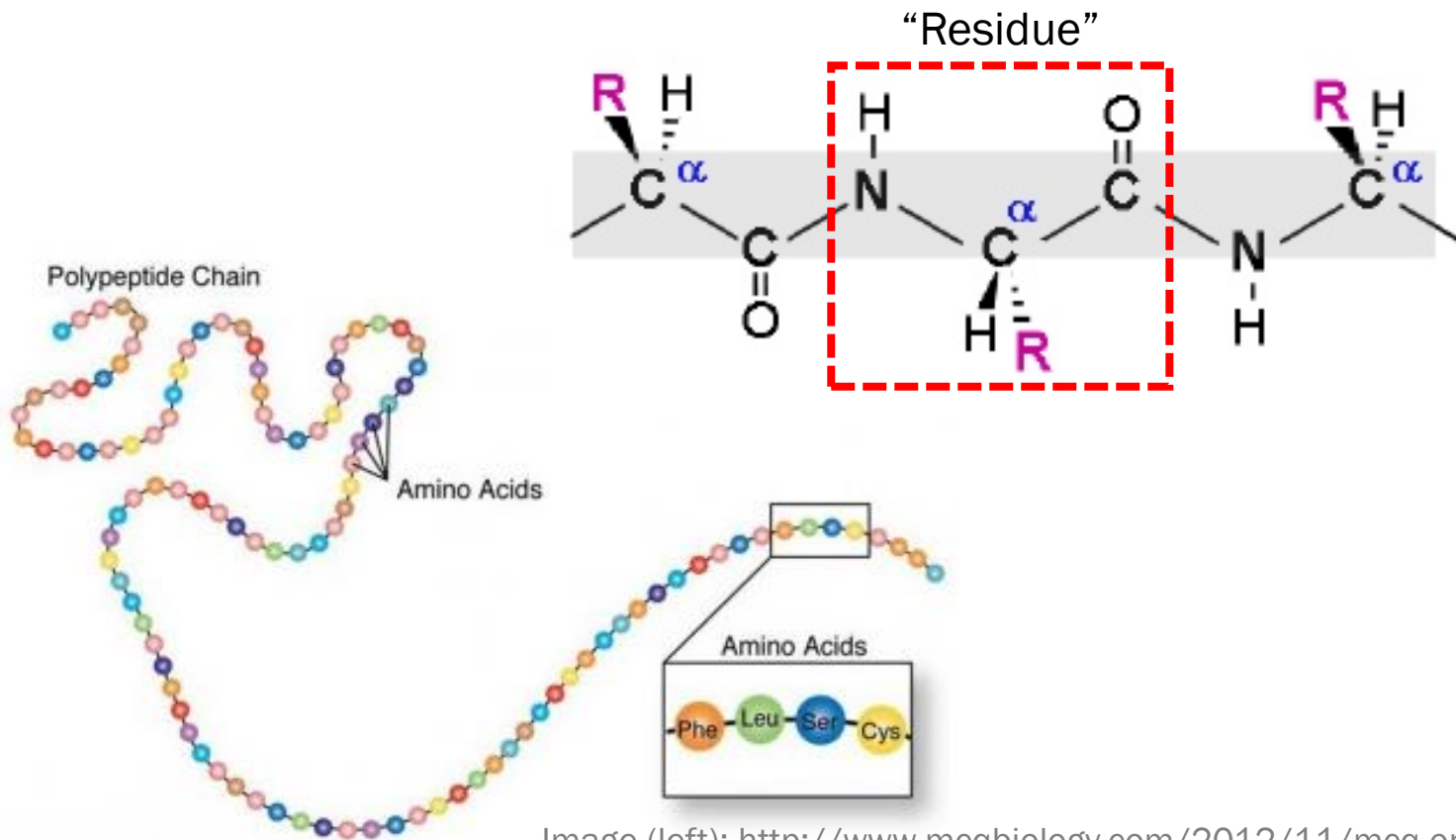


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>

Proteins are Polymer Chains

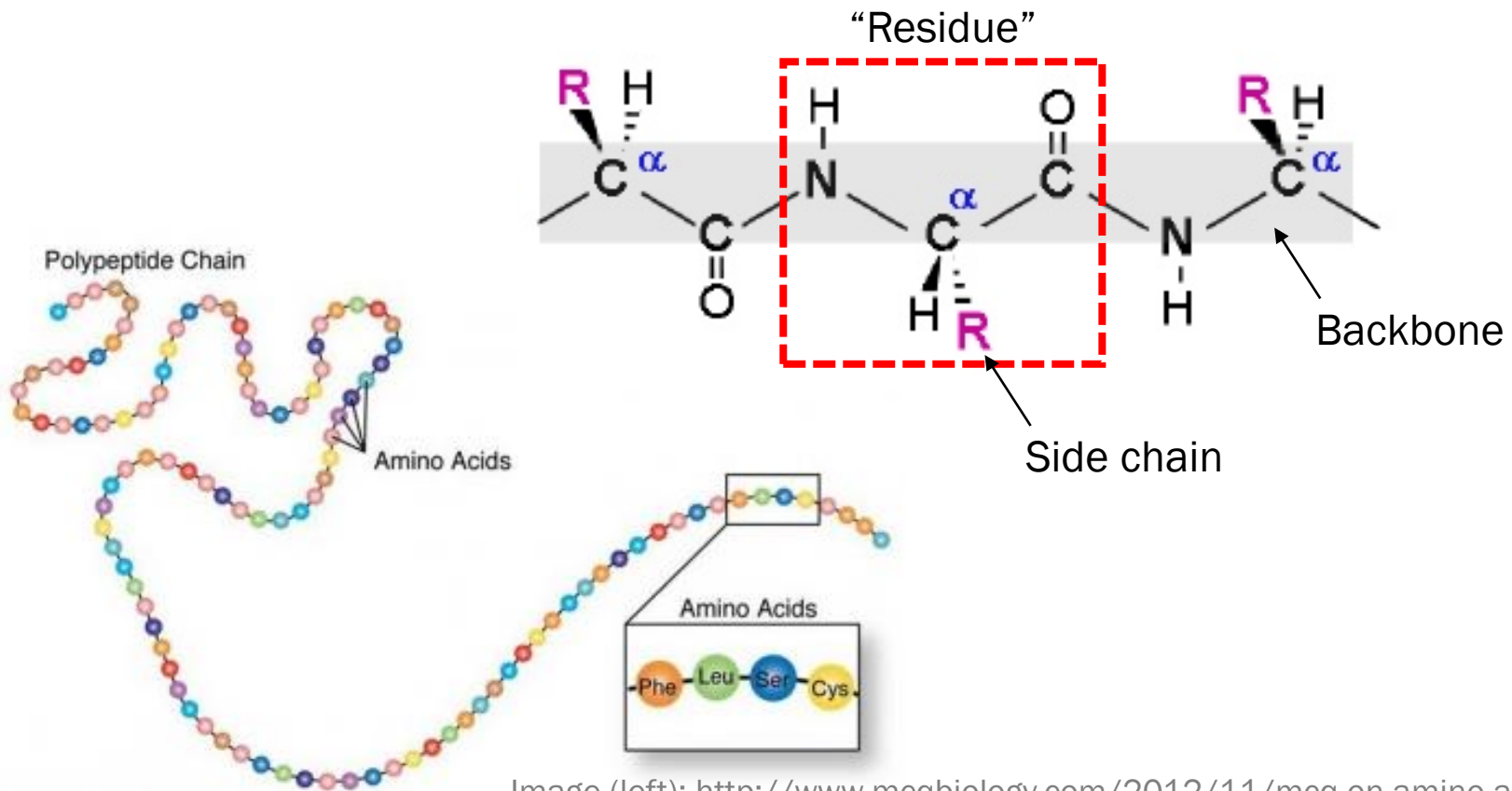


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

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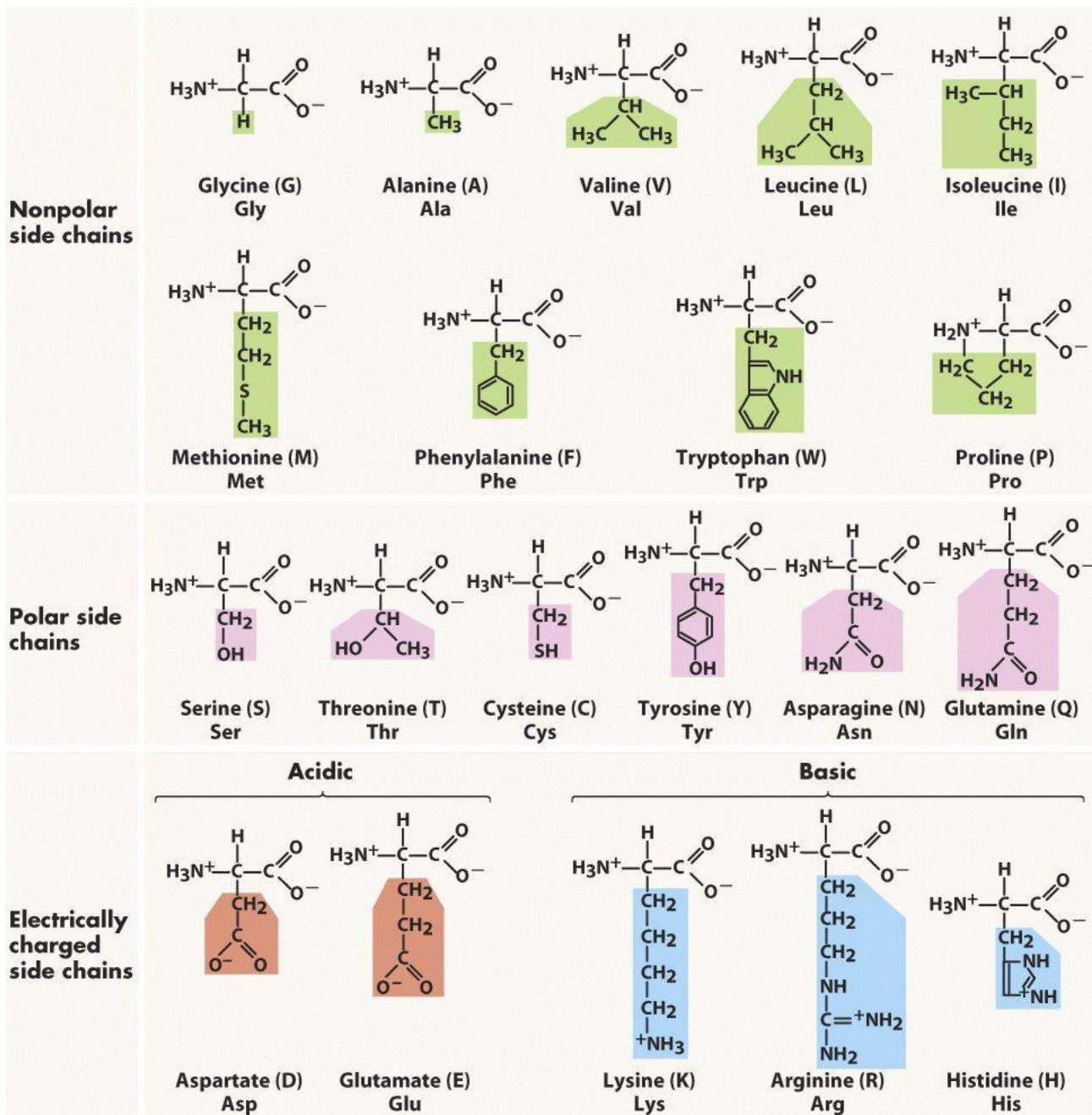
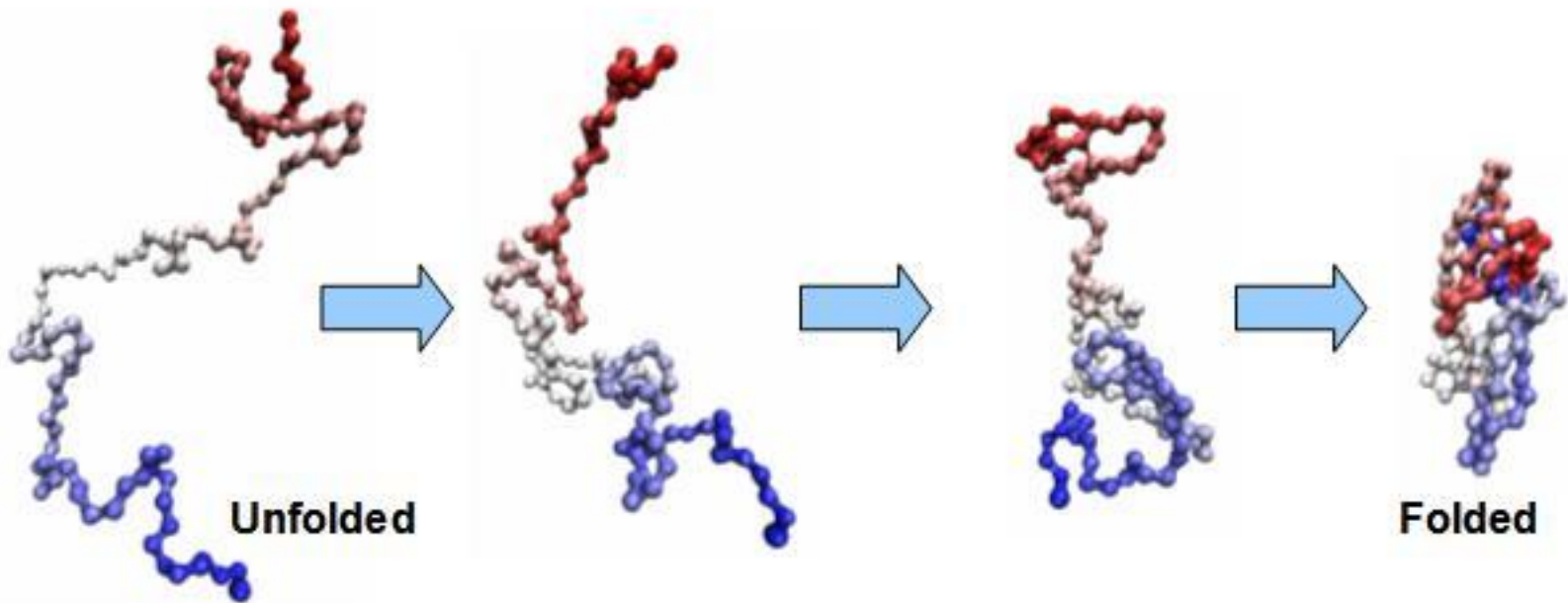
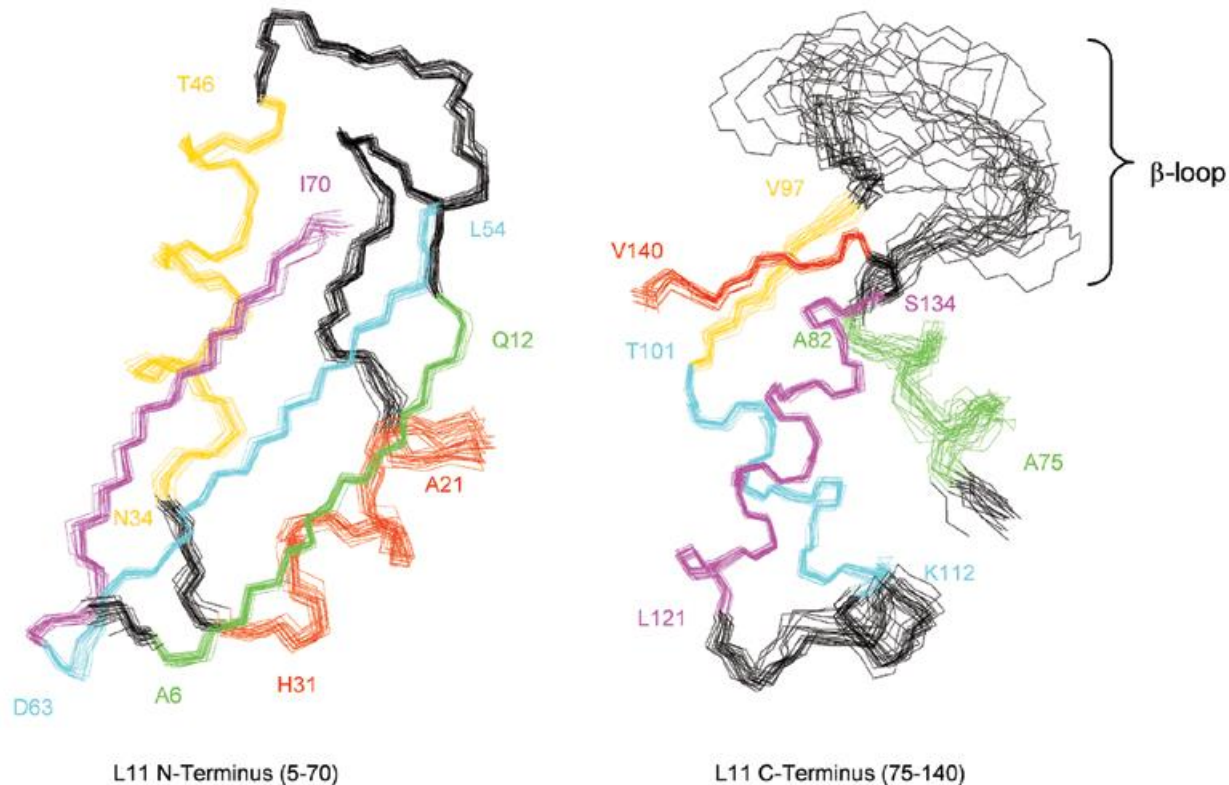


Figure 3-5 Biological Science, 2/e

Canonical Proteins Require Folding



“Frozen” in the Folded State



(measurement by NMR)

Image: <http://schwalbe.org.chemie.uni-frankfurt.de/node/684>

Thermodynamics of Folding

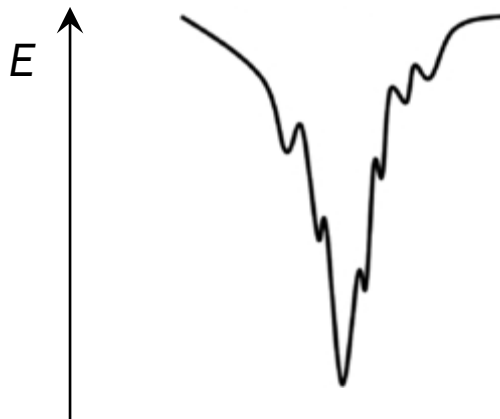


Image (top): <https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118>

Image (bottom): Mészáros et al., *Phys Biol* (2011)

What If ...

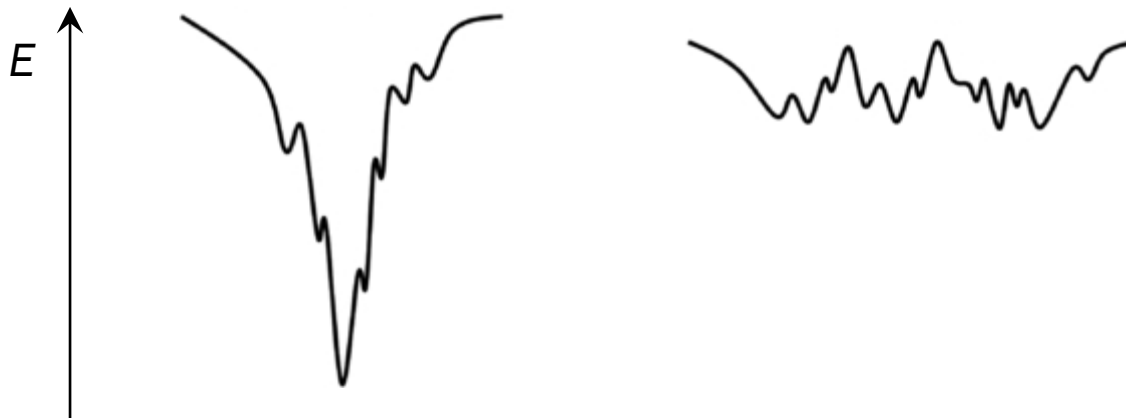


Image (top): <https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118>

Image (bottom): Mészáros et al., *Phys Biol* (2011)

No Fixed Structure: Flexibility!

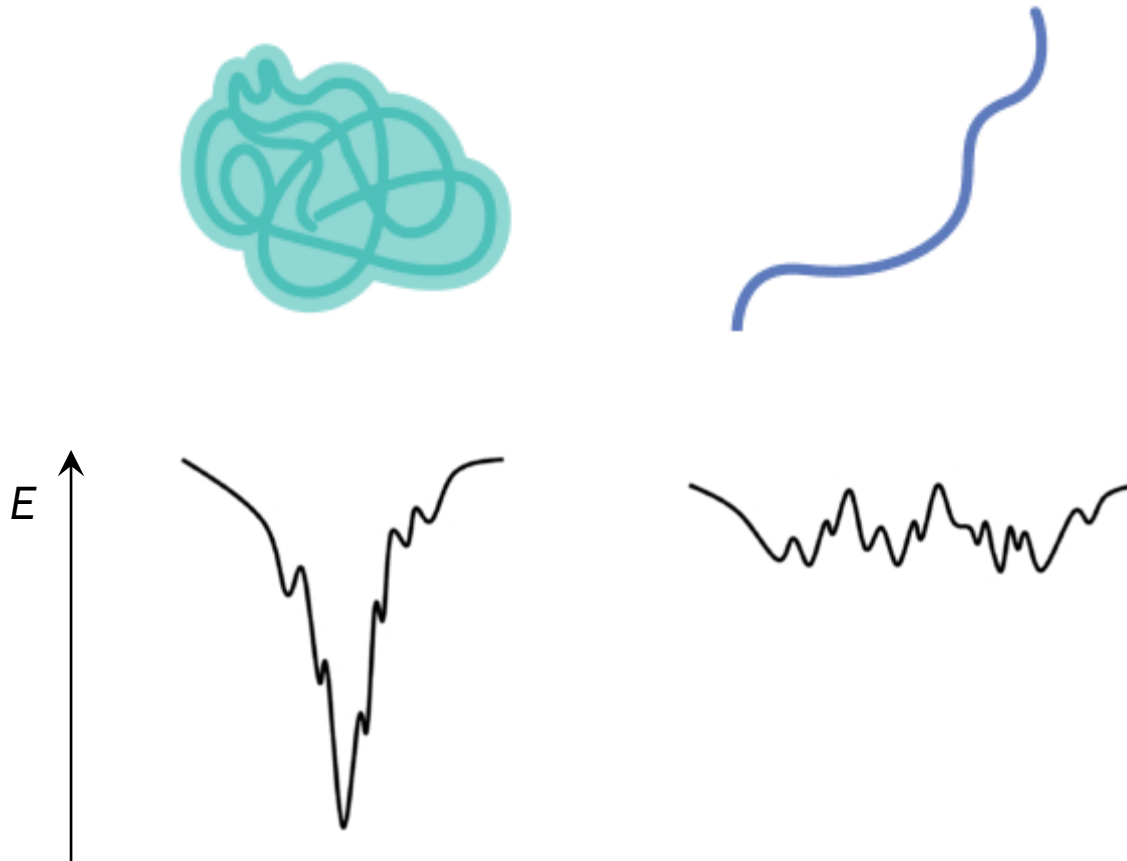


Image (top): <https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118>

Image (bottom): Mészáros et al., *Phys Biol* (2011)

Intrinsically Disordered Protein (IDP)



foldable protein



intrinsically disordered protein

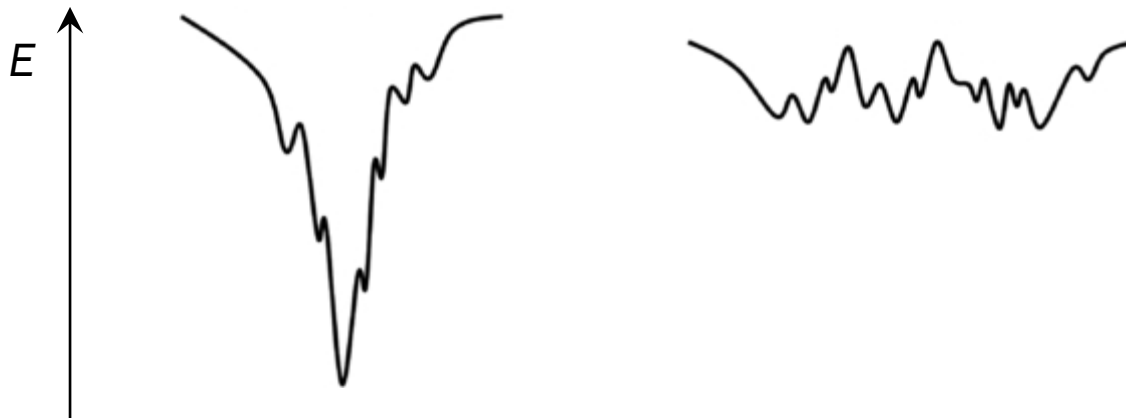
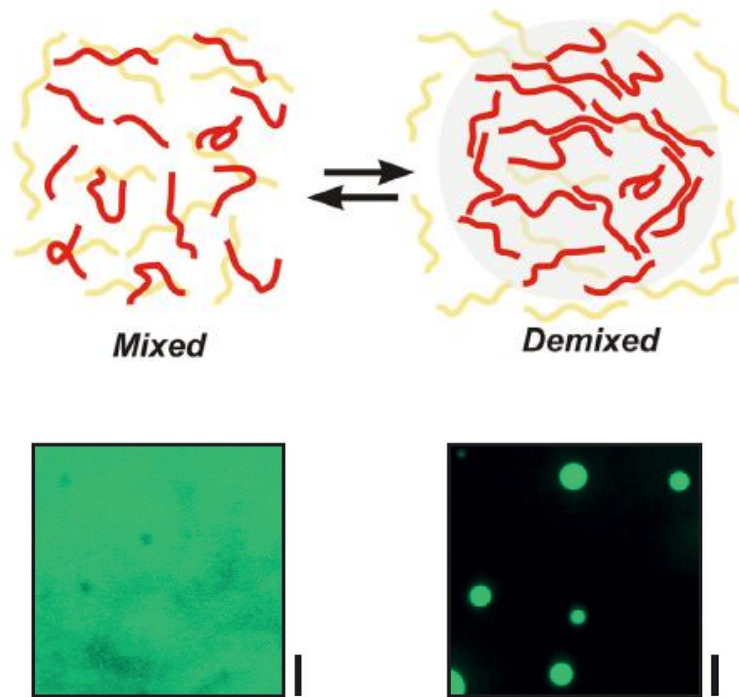


Image (top): <https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118>

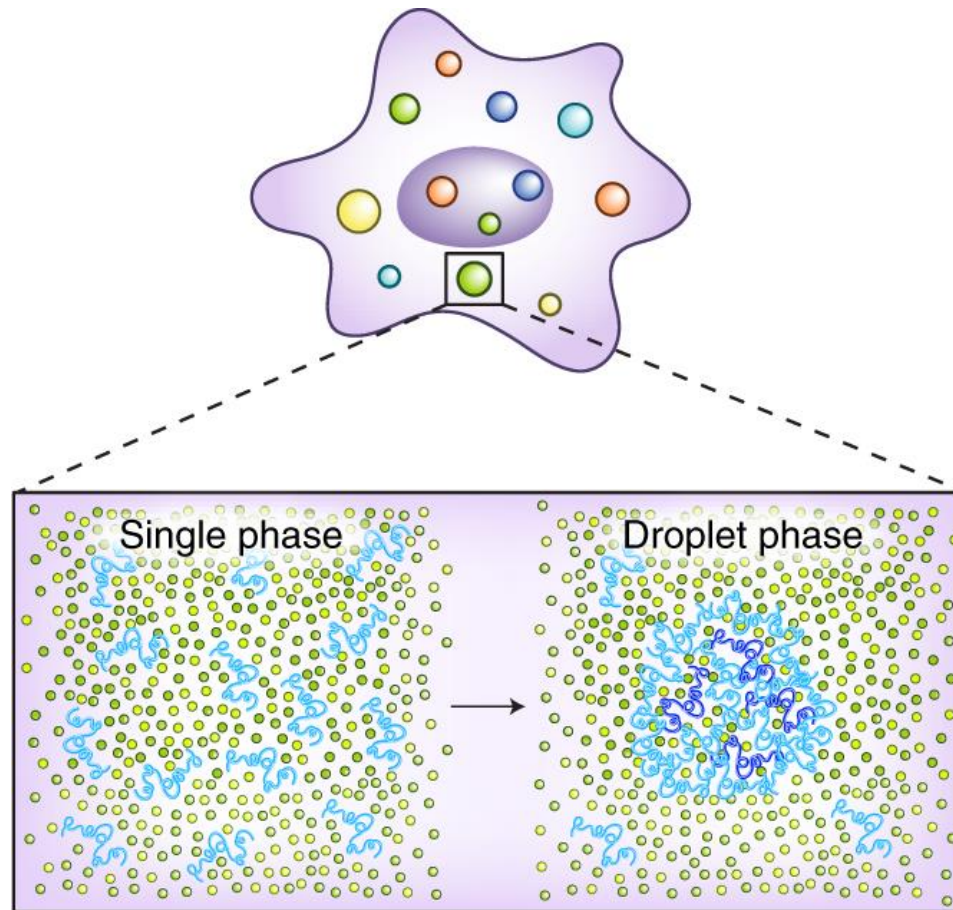
Image (bottom): Mészáros et al., *Phys Biol* (2011)

IDPs and Phase Separation

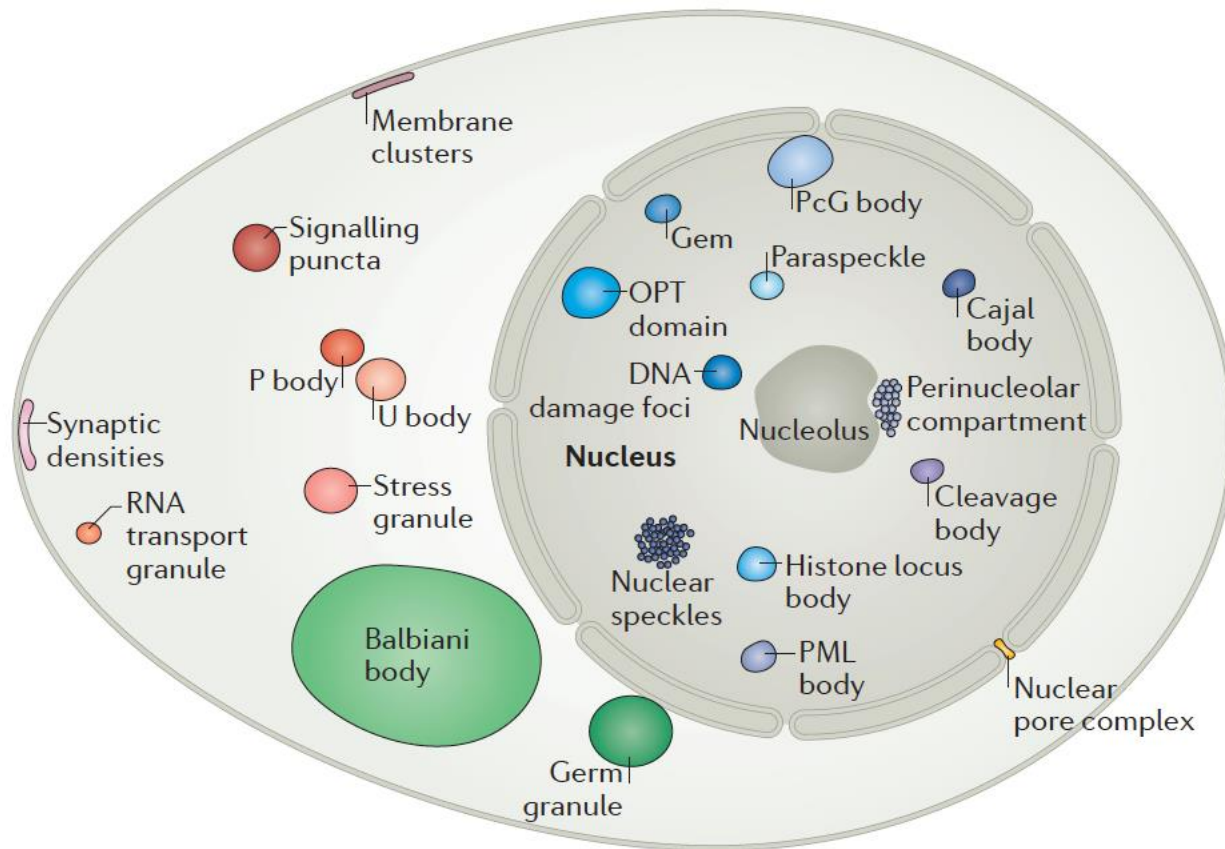


(Scale bars: 5 μm)

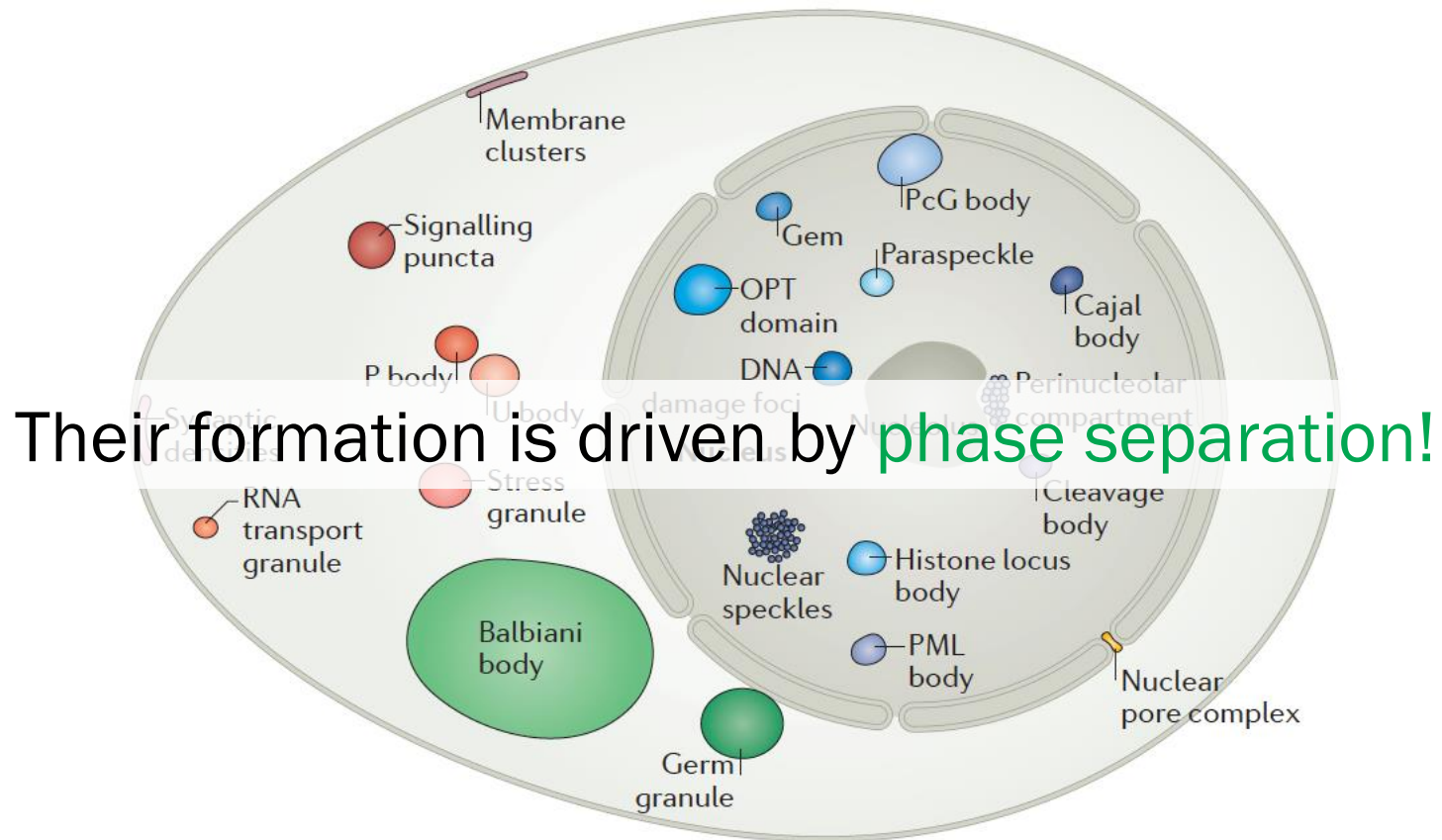
Intracellular Phase Separation



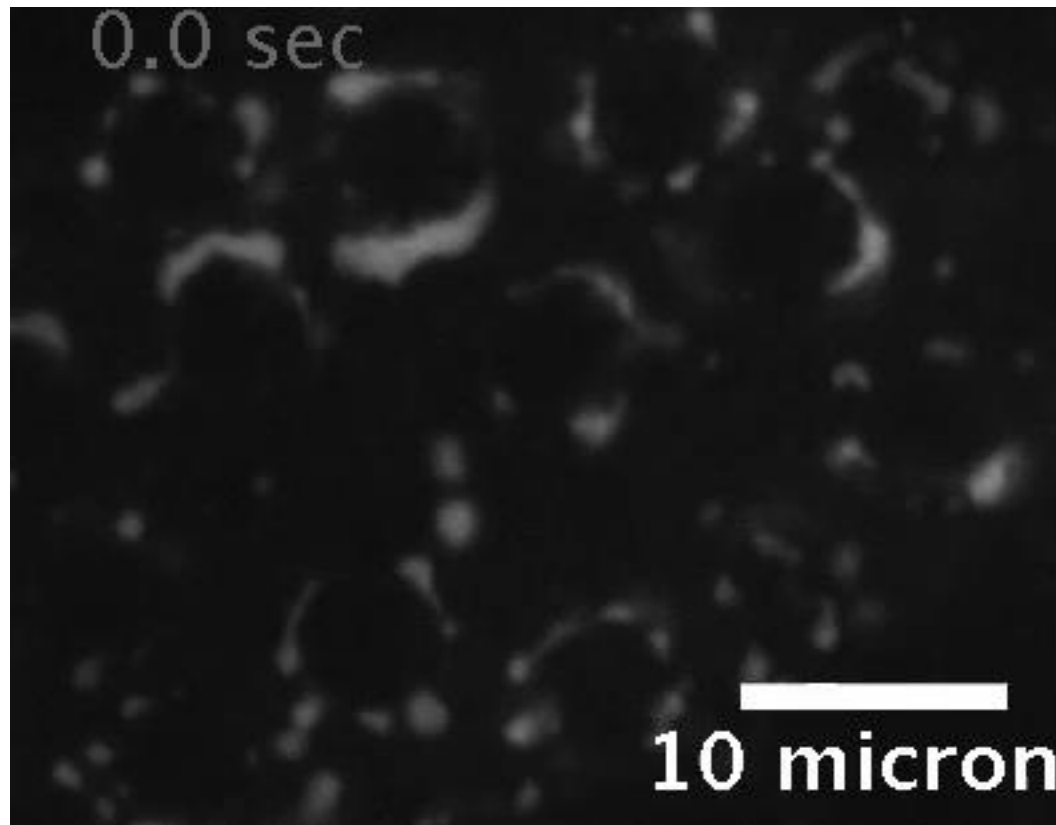
Membraneless Organelles (MLOs)



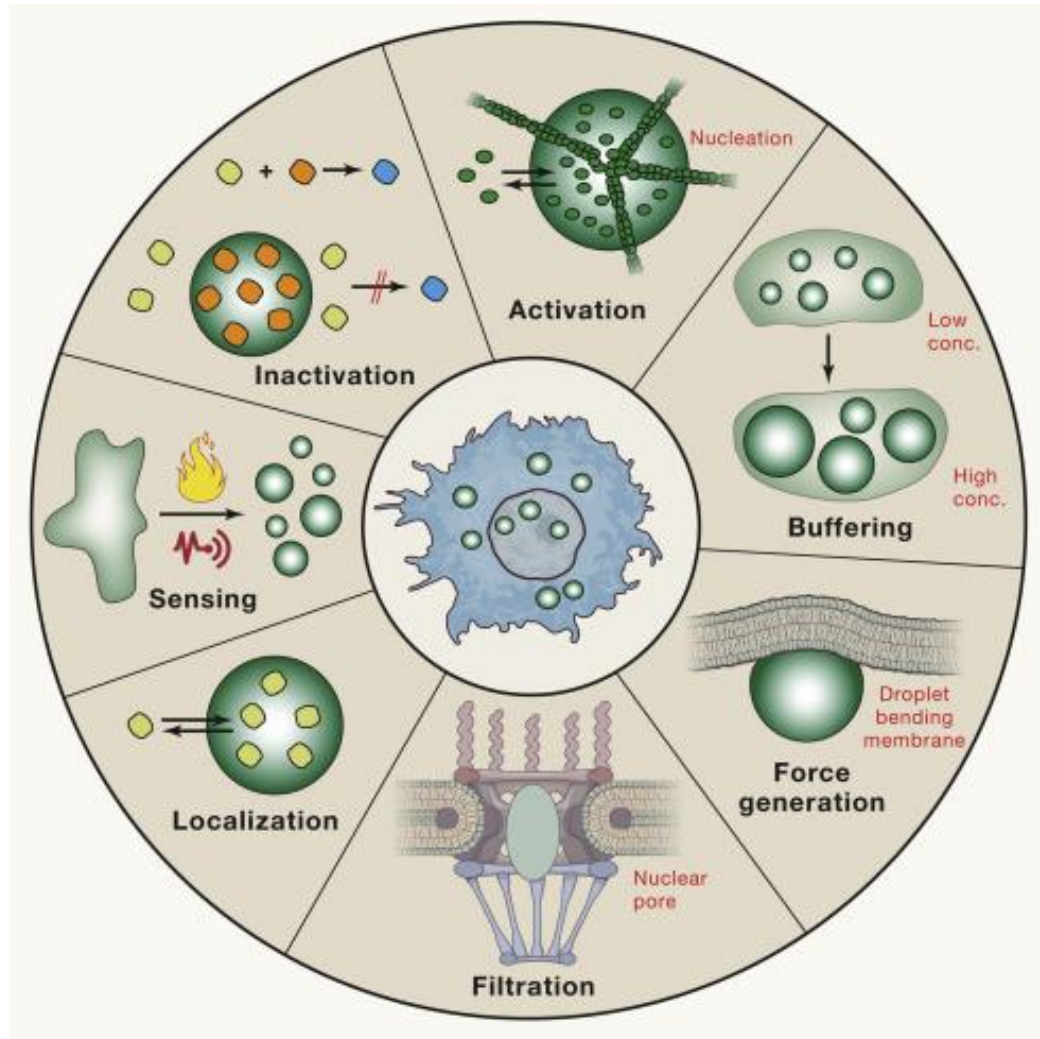
Membraneless Organelles (MLOs)



Liquidity of MLOs (even *in vivo*)

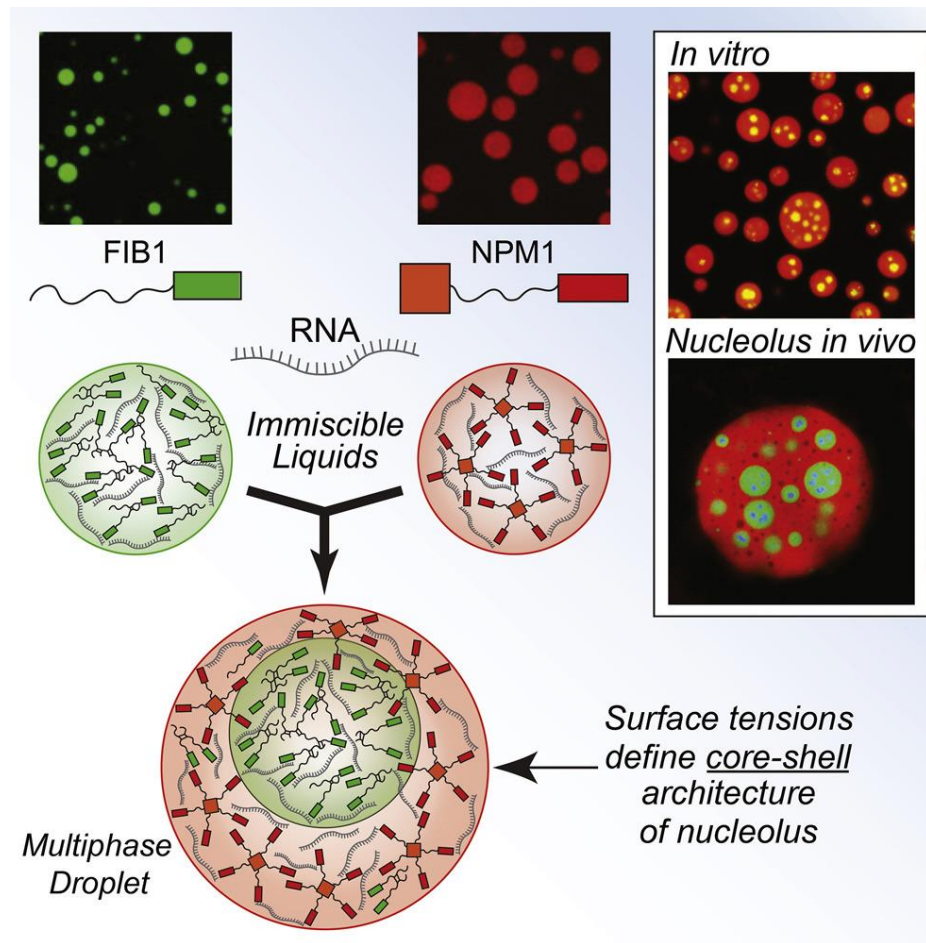


Biological Roles of MLOs

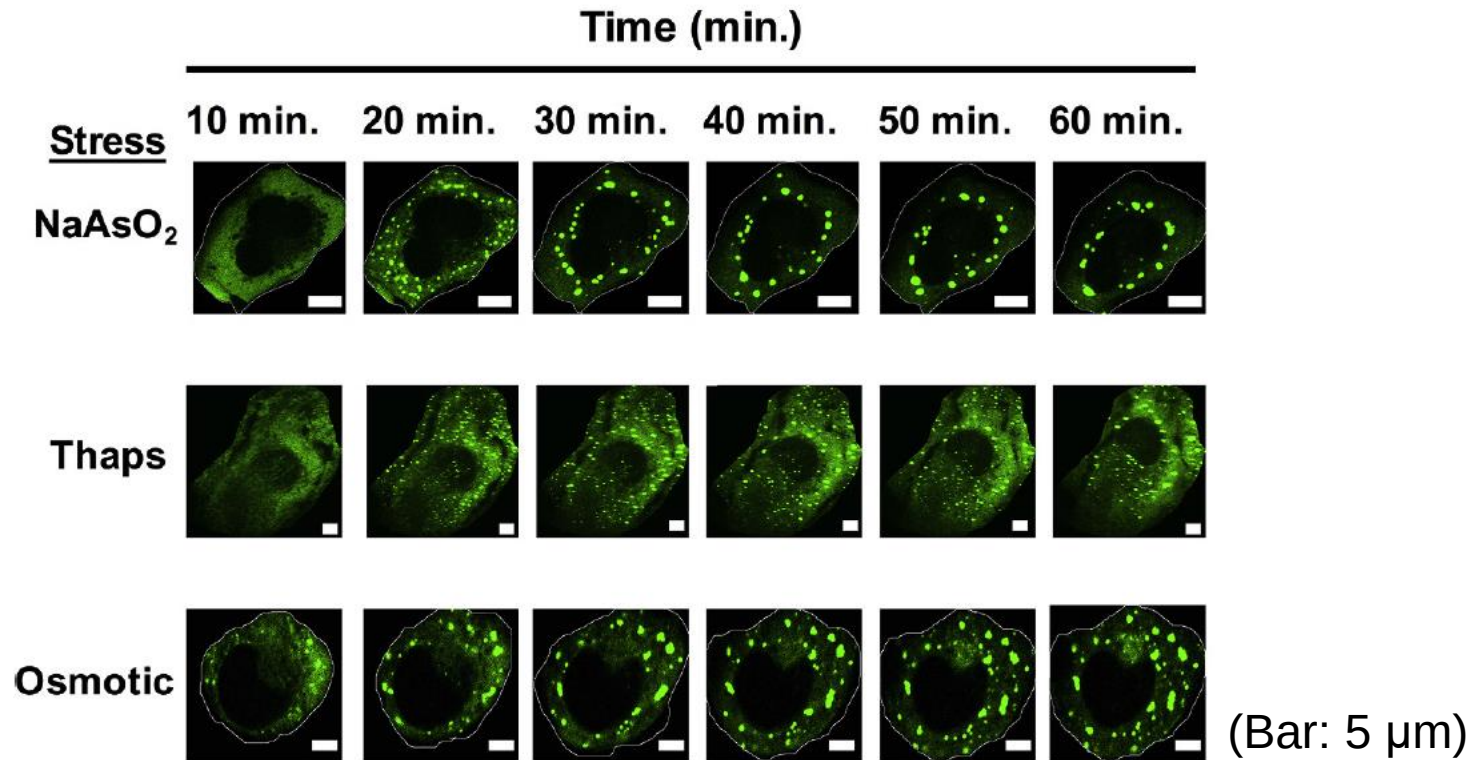


Alberti et al., *Cell* (2019)

Example: Nucleolus Formation



Example: Stress Granules



human osteosarcoma cells (U-2 OS)
Signal from GFP-G3BP1 fusion protein

Wheeler et al., *eLife* (2016)

LLPS and Neuroscience

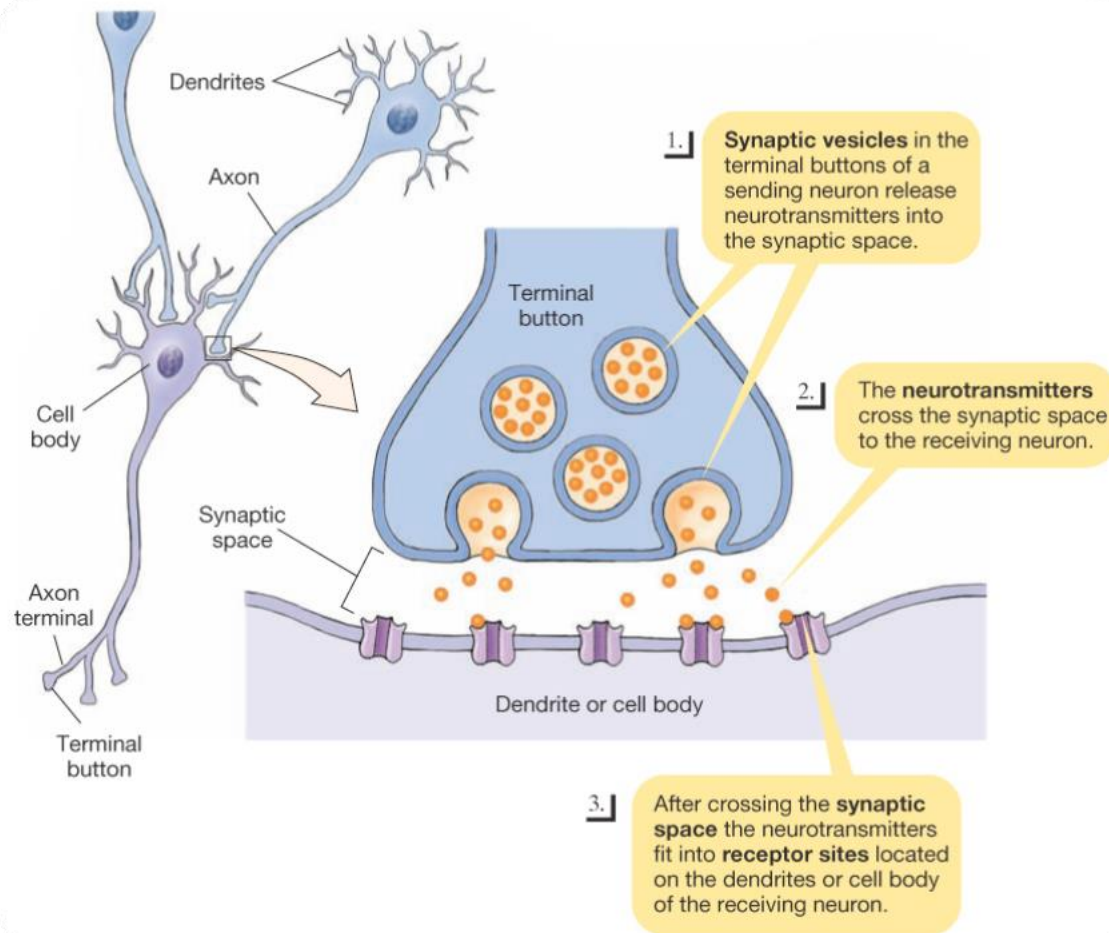
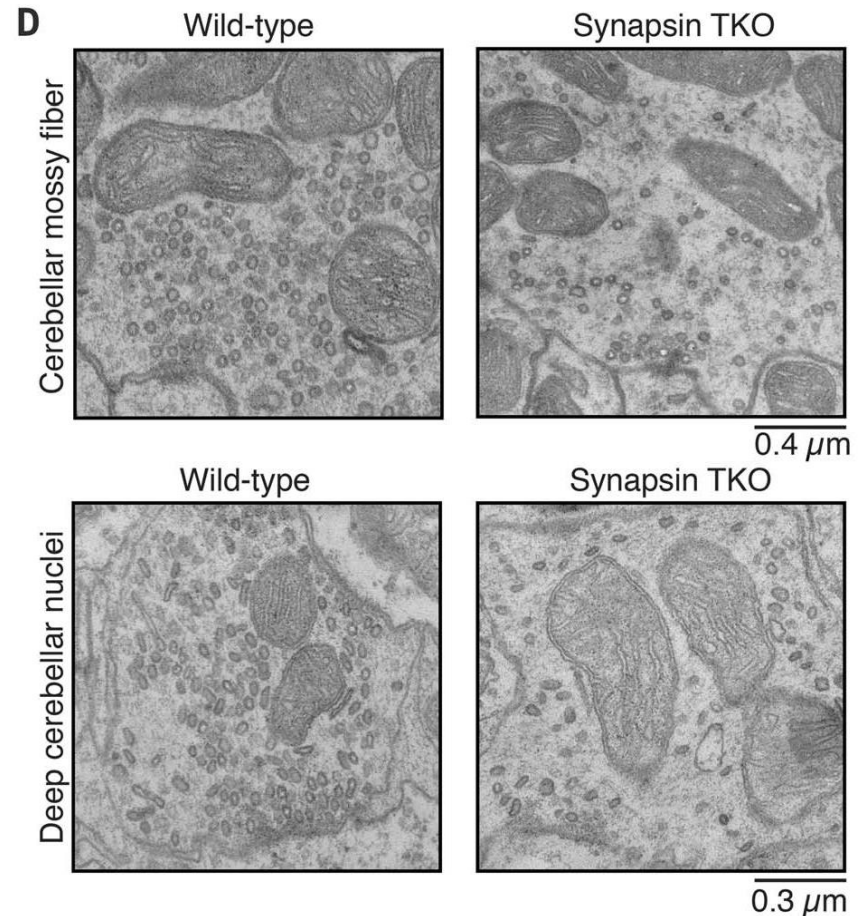
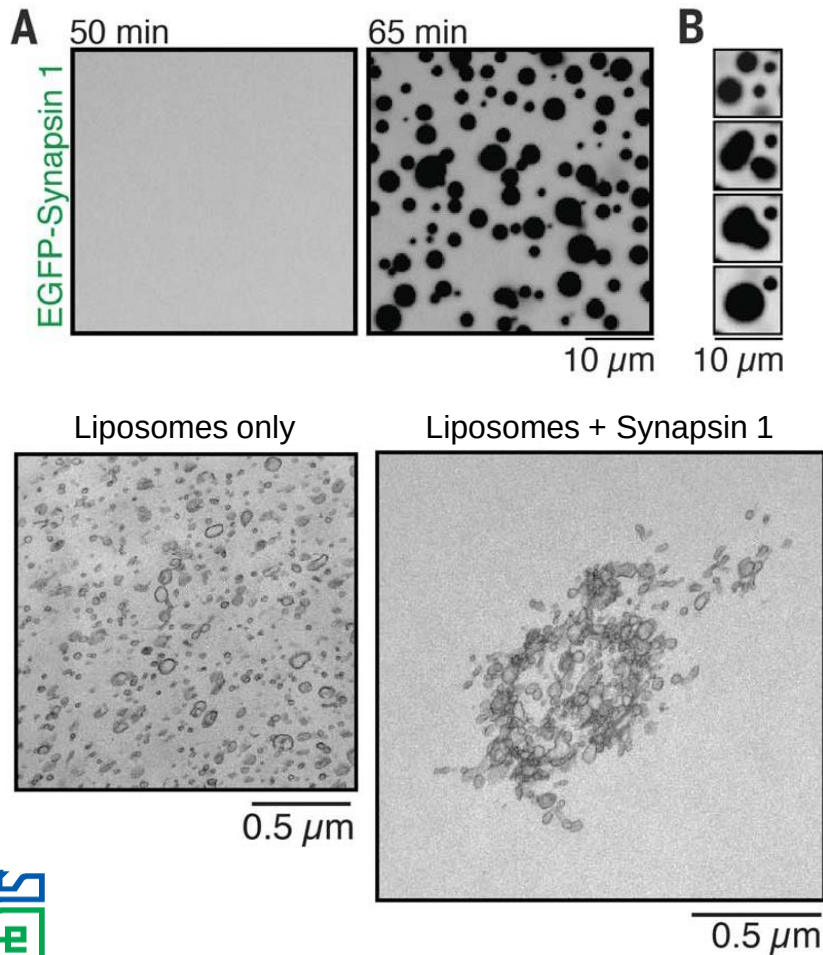


Image: <https://socratic.org/questions/how-do-impulses-travel-across-a-synapse>

LLPS and Neuroscience



Milovanovic et al., *Science* (2018)

Neurodegenerative Diseases

Alzheimer's disease

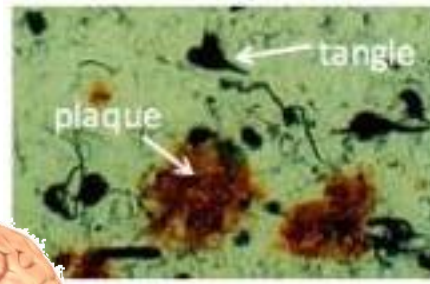
amyloid beta, tau

Parkinson's disease

α -synuclein

Huntington's disease

huntingtin



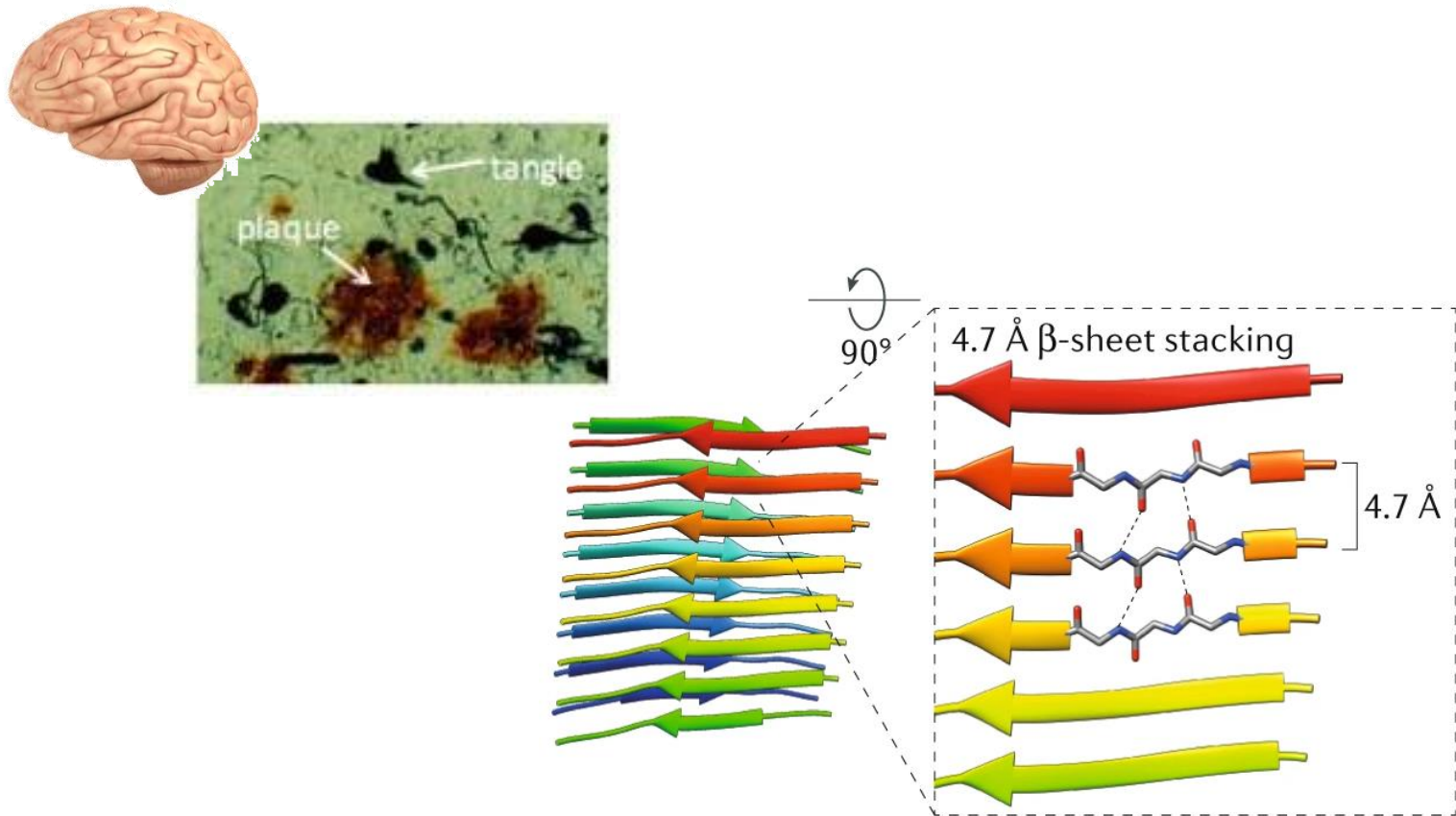
BSE (mad cow disease)

prion

ALS (Lou Gehrig's disease)

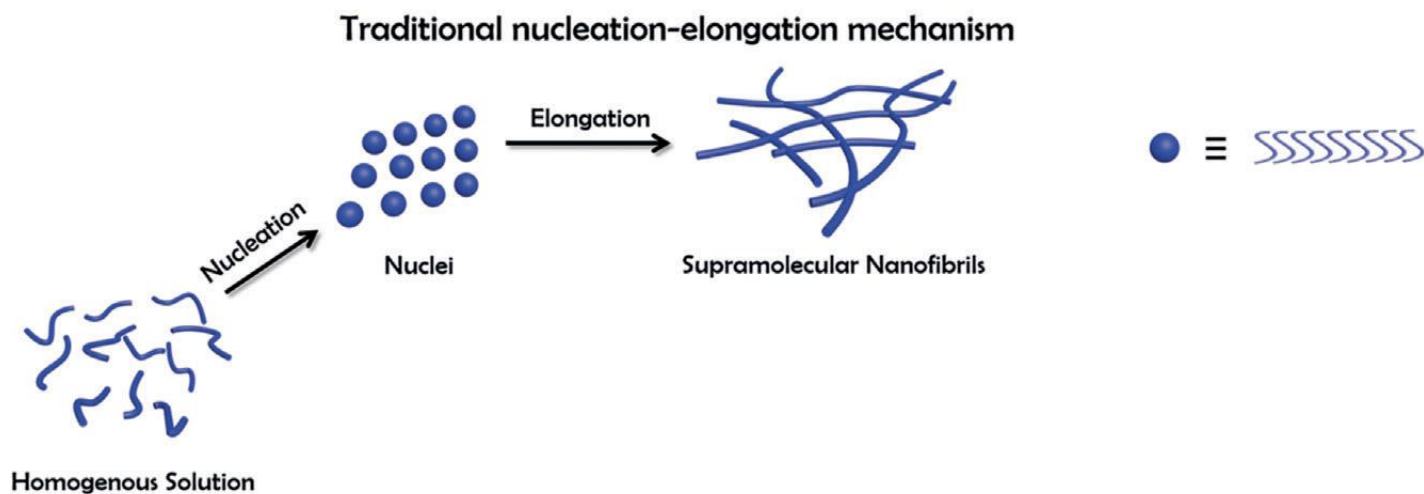
TDP43, FUS

Deposition of Amyloid Fibrils

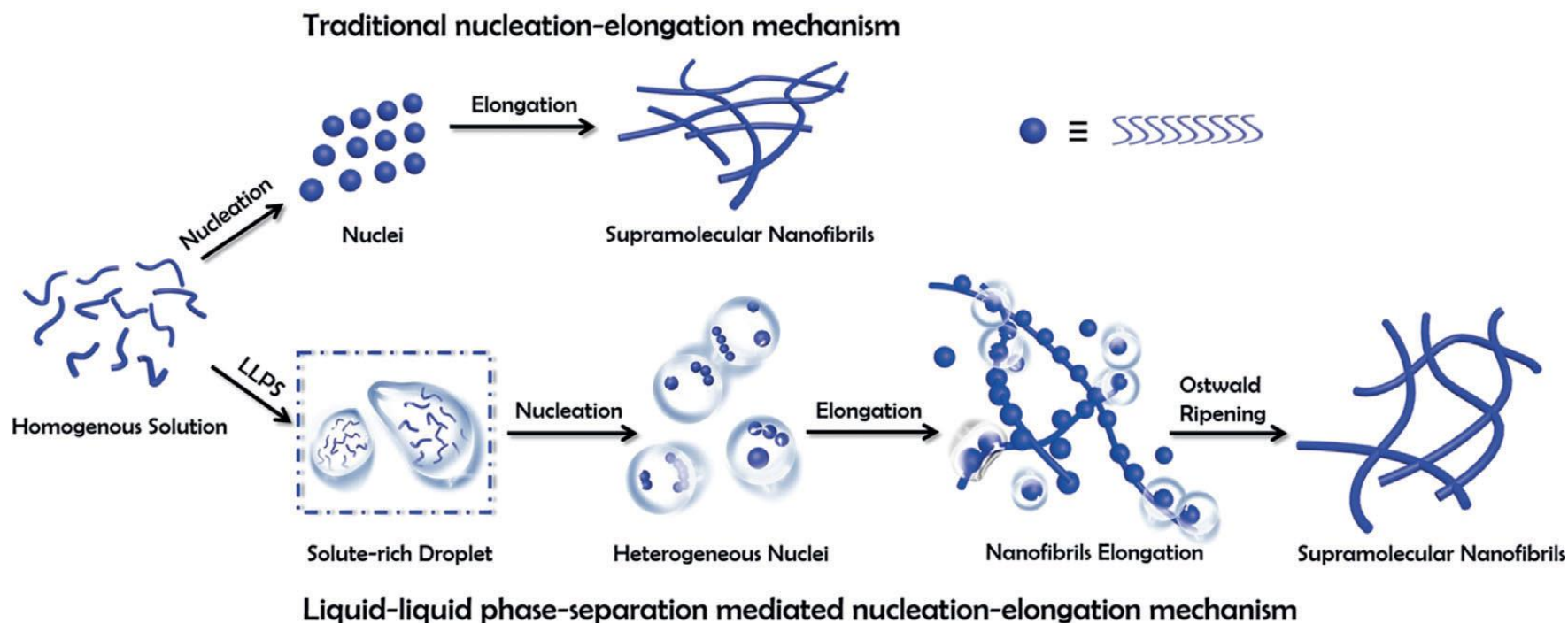


Iadanza et al., *Nat Rev Mol Cell Biol* (2018)

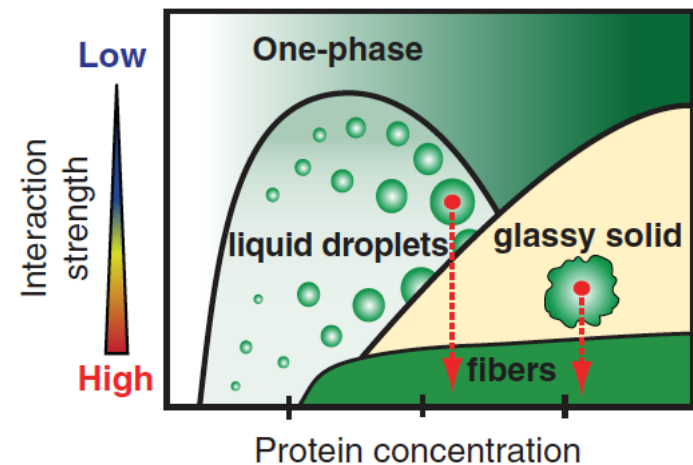
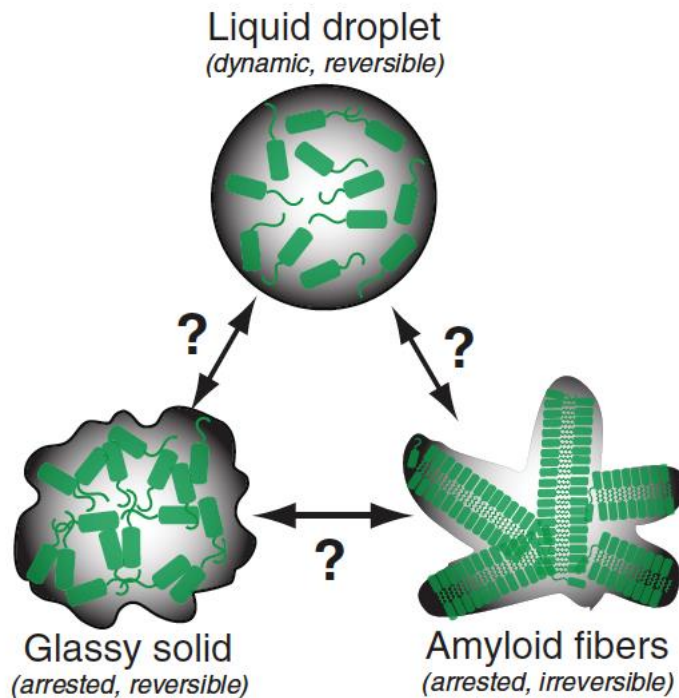
Traditional Hypothesis



Another Route: via Phase Separation

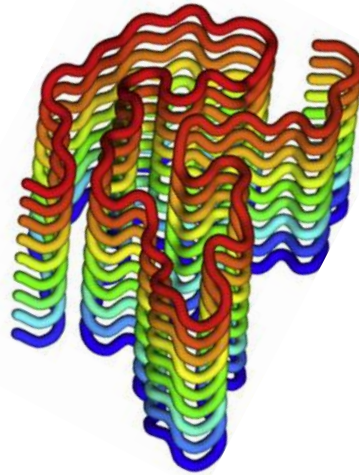
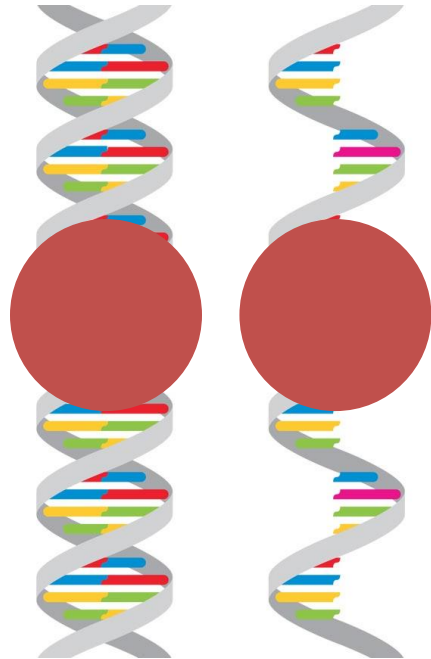


IDPs Show Rich Phase Behaviors

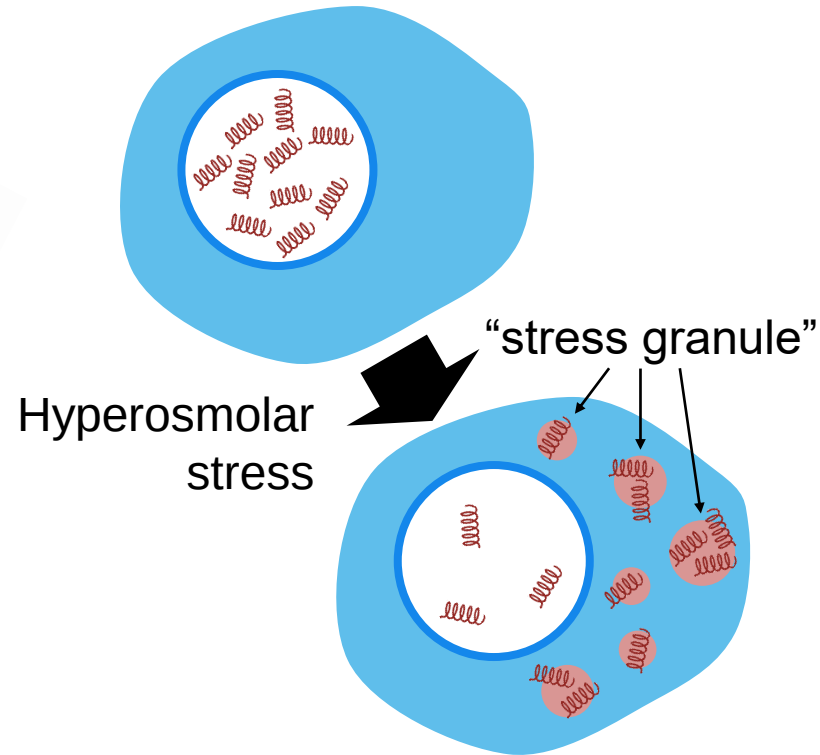


FUS (*Fused in Sarcoma*) Protein

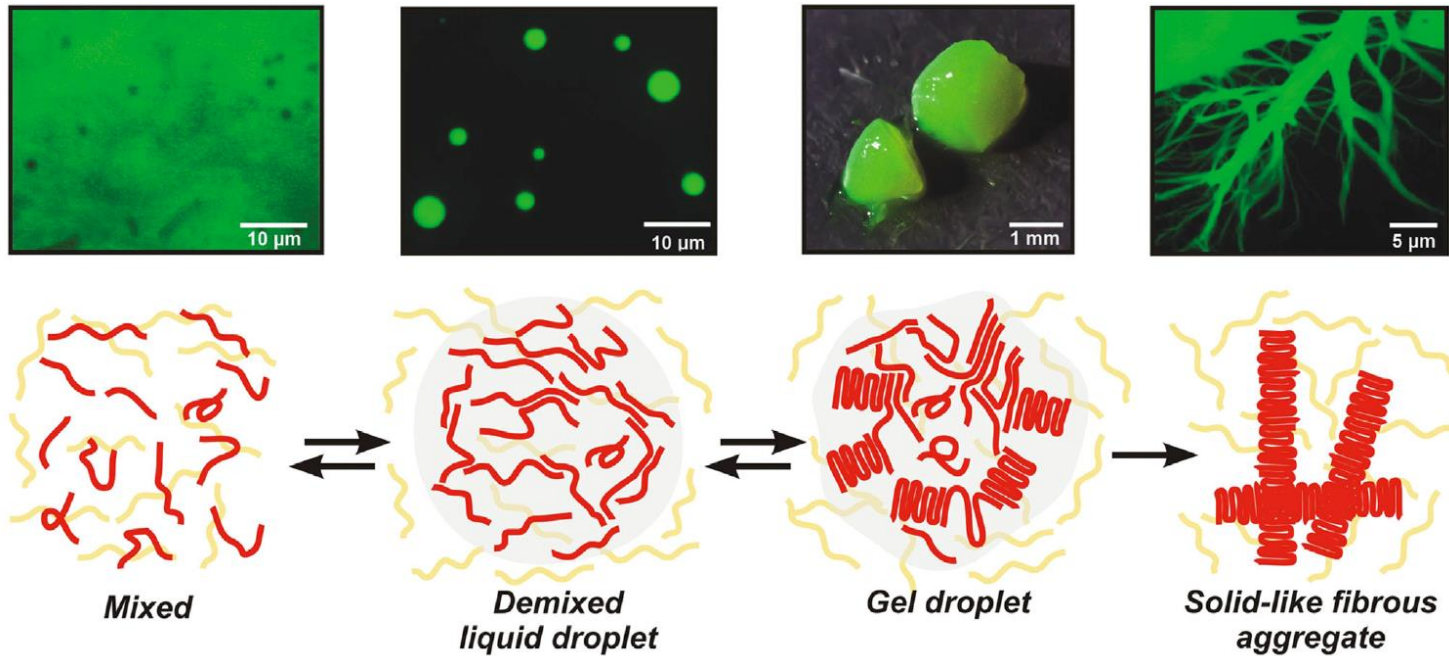
DNA/RNA binding Fibril formation in ALS



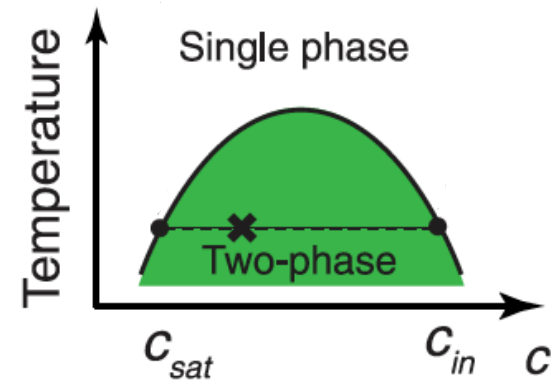
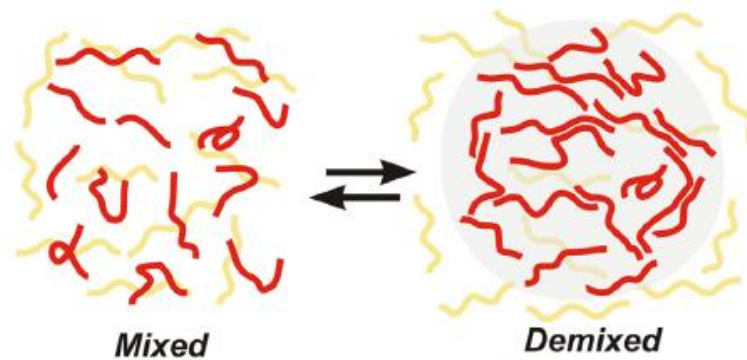
Stress response



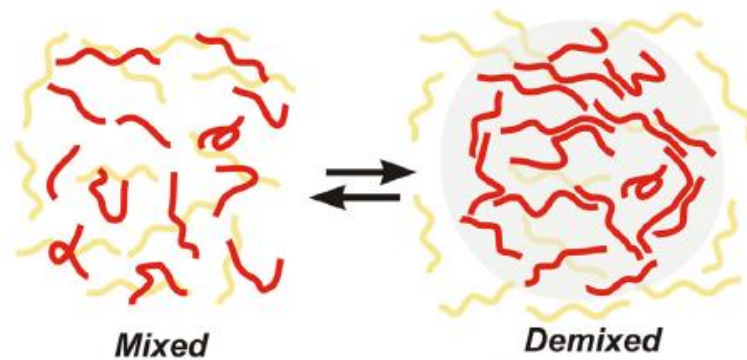
Phase Behaviors of FUS



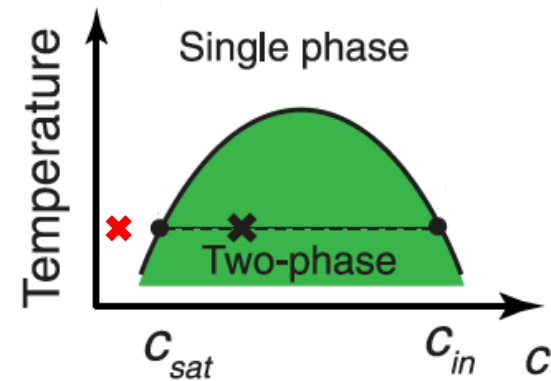
Readout for Phase Behavior



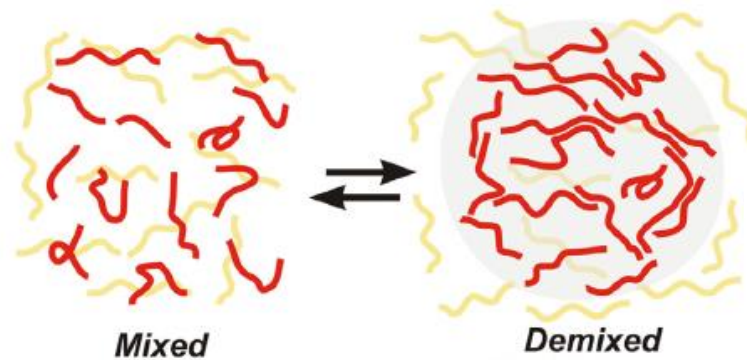
Readout for Phase Behavior



1 μ M

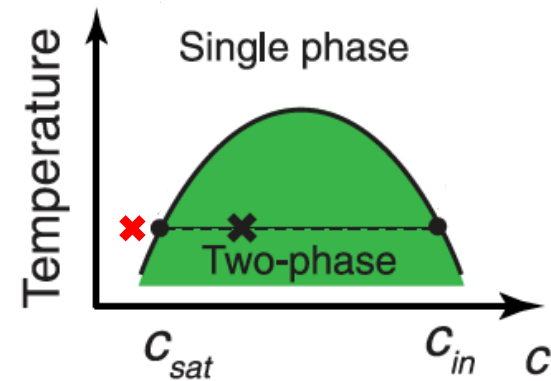
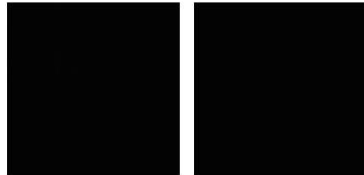


Readout for Phase Behavior

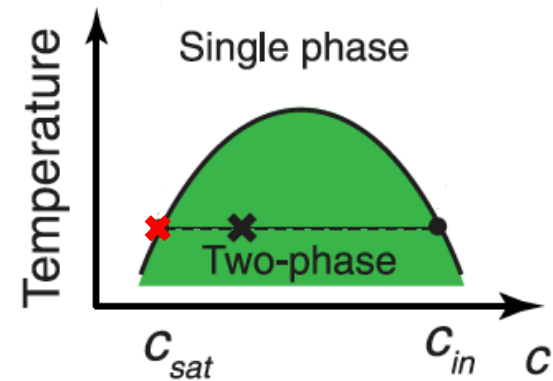
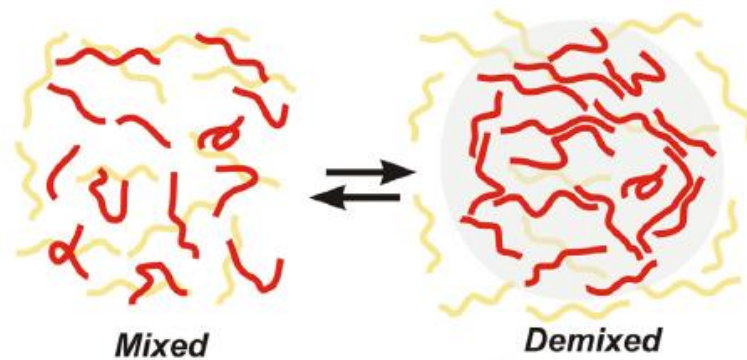


1 μM

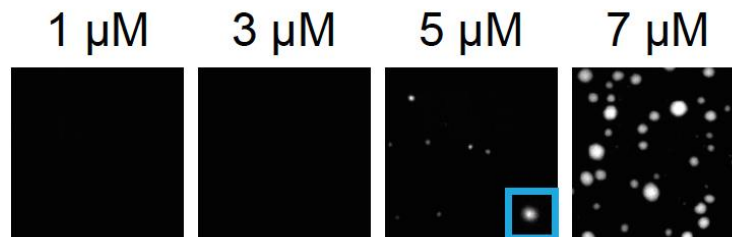
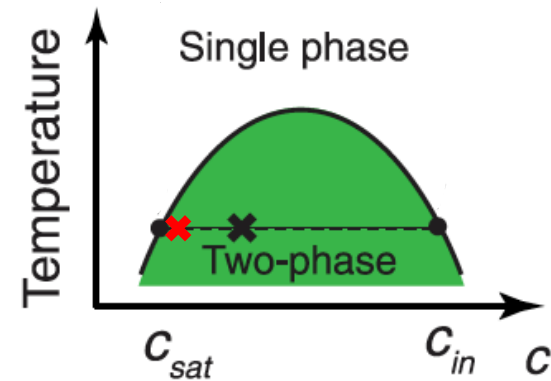
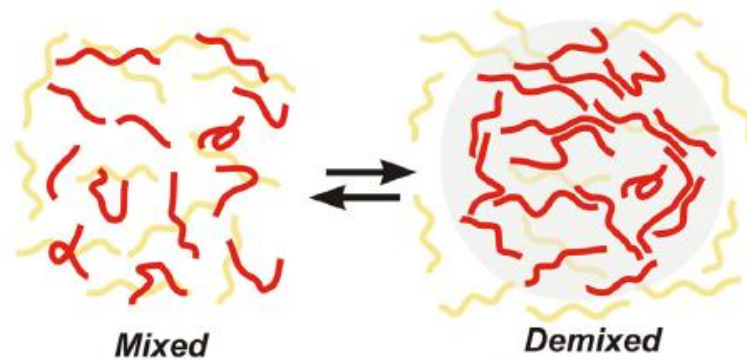
3 μM



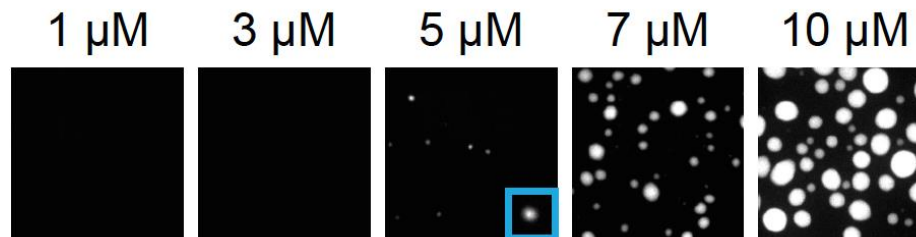
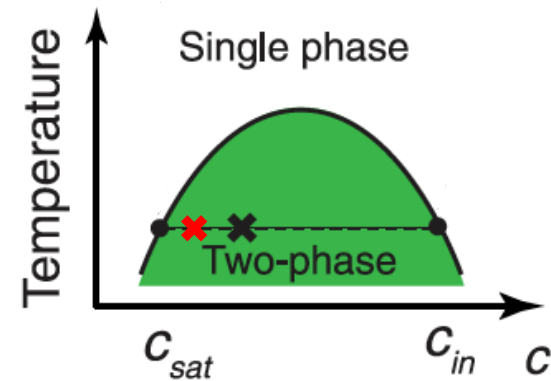
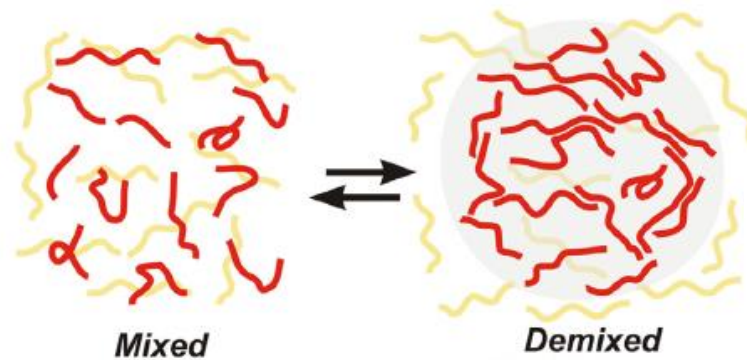
Readout for Phase Behavior



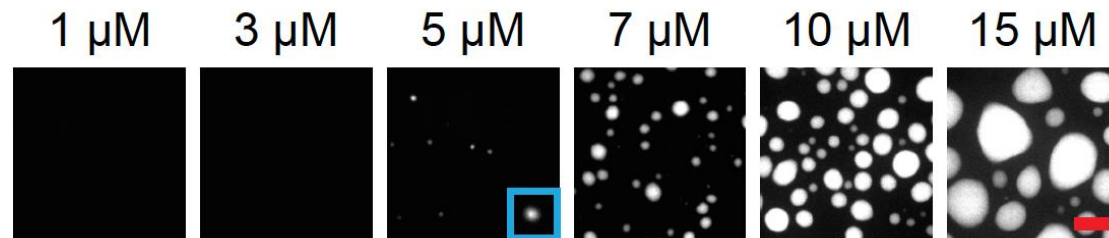
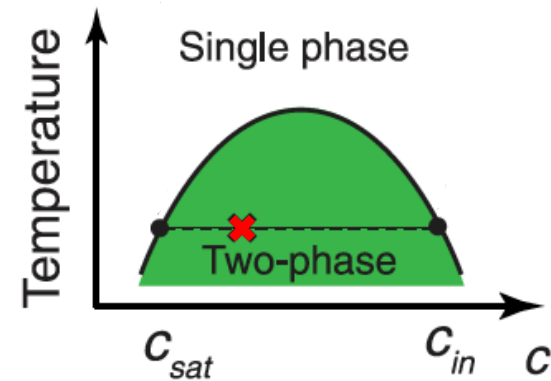
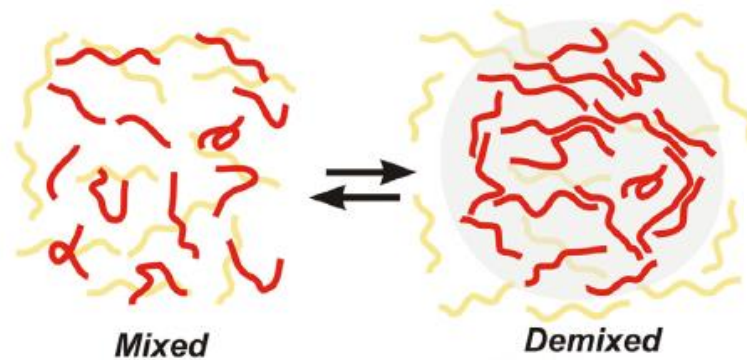
Readout for Phase Behavior



Readout for Phase Behavior

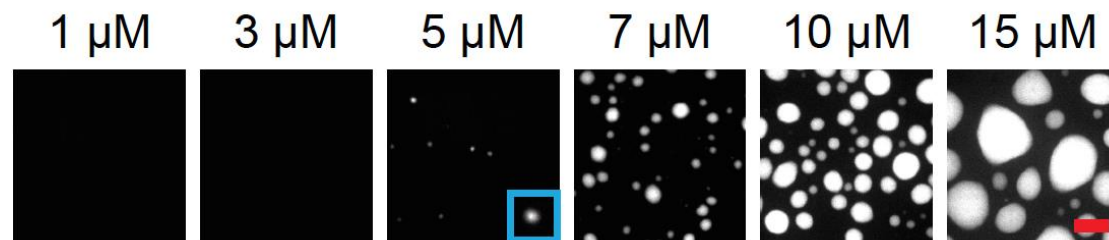
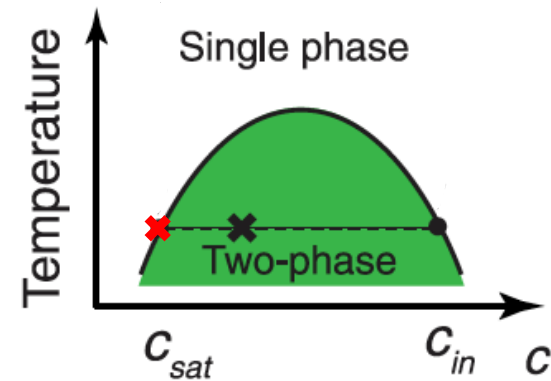
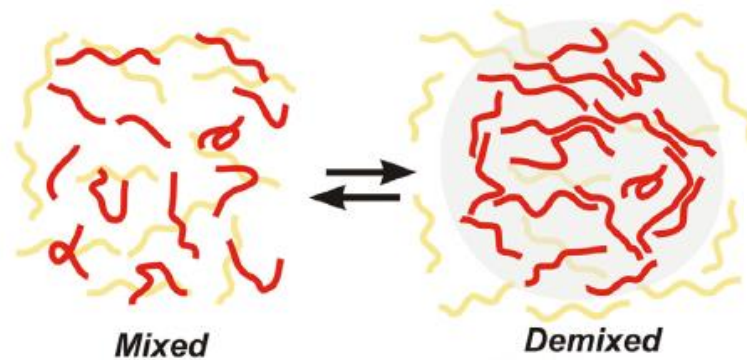


Readout for Phase Behavior



Bar: 5 μm

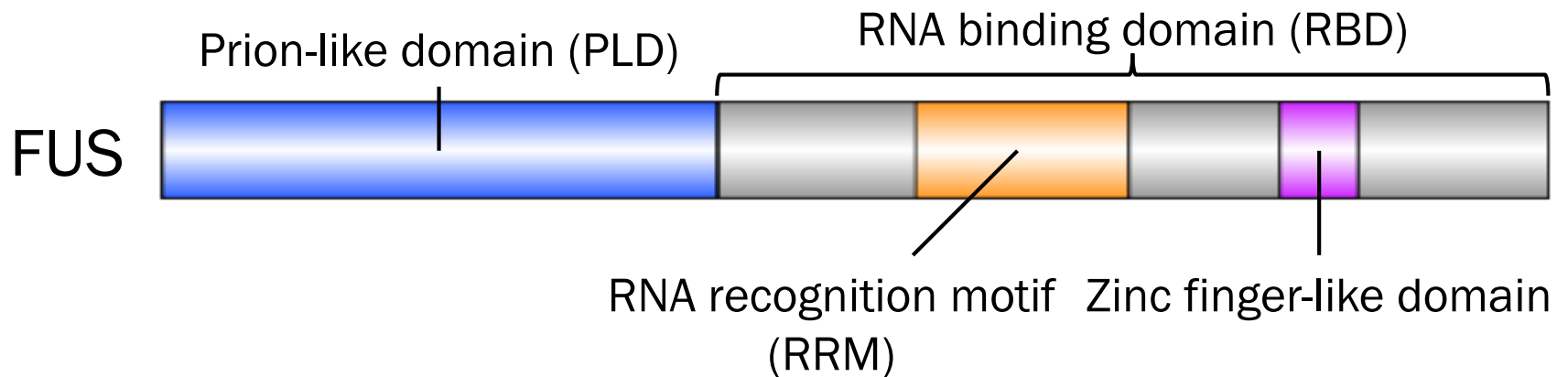
Readout for Phase Behavior



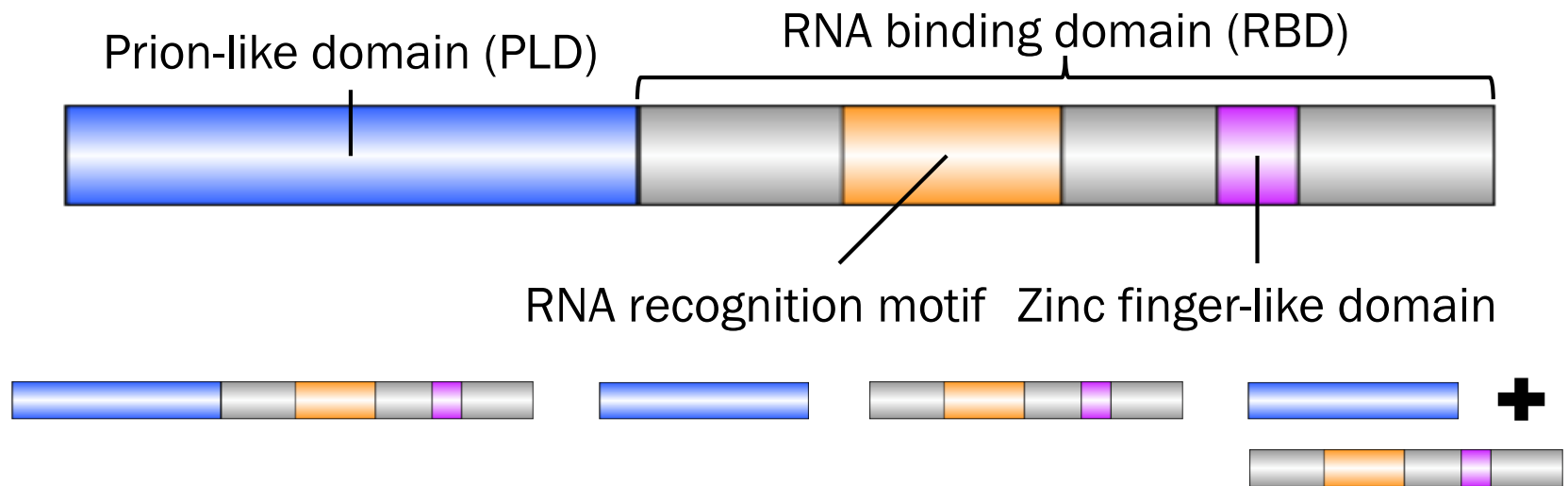
Bar: 5 μm

saturation concentration c_{sat}

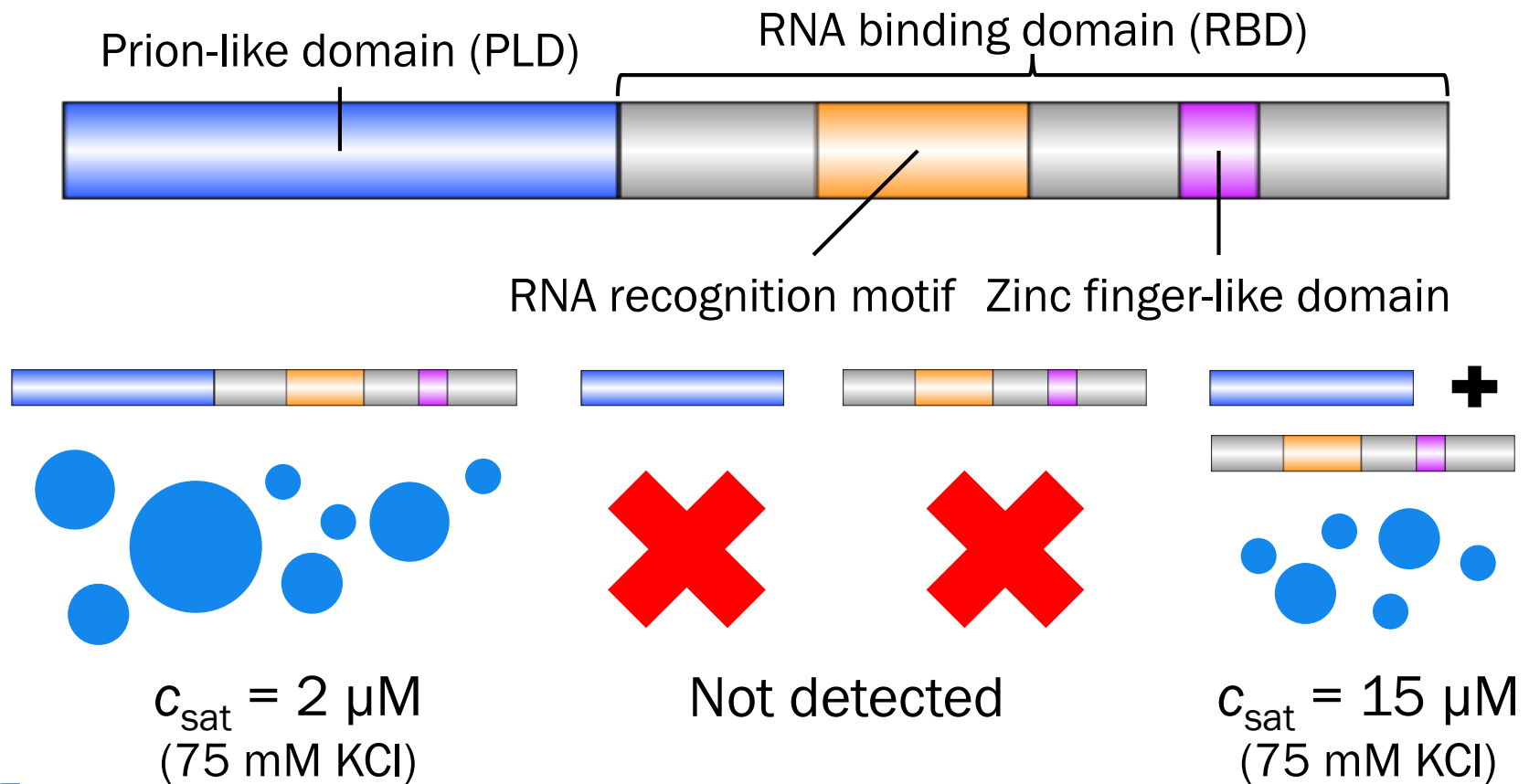
Domain Architecture of FUS



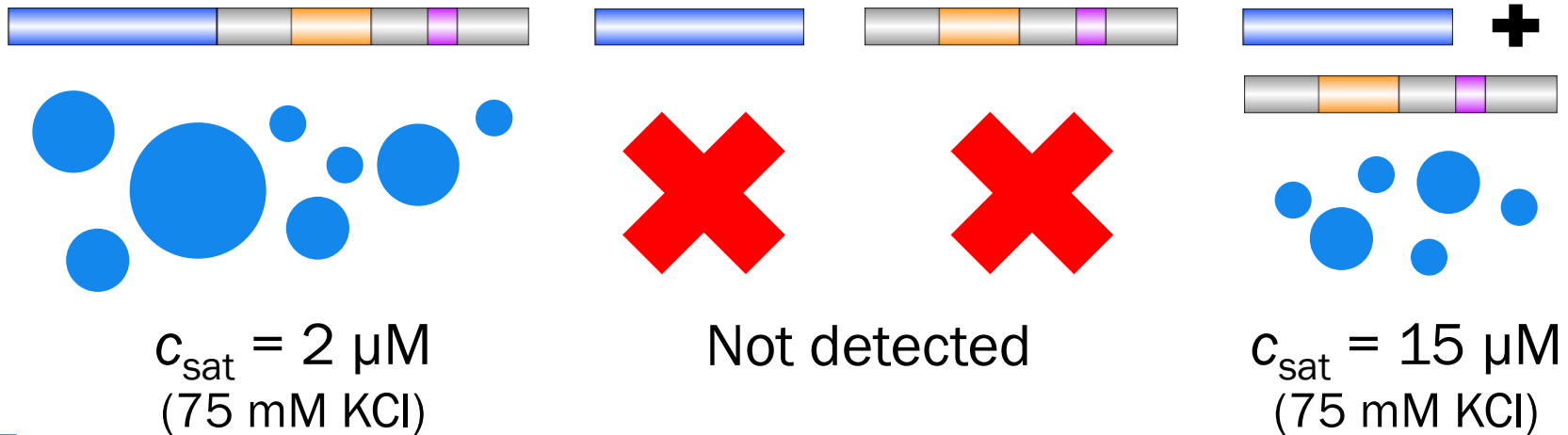
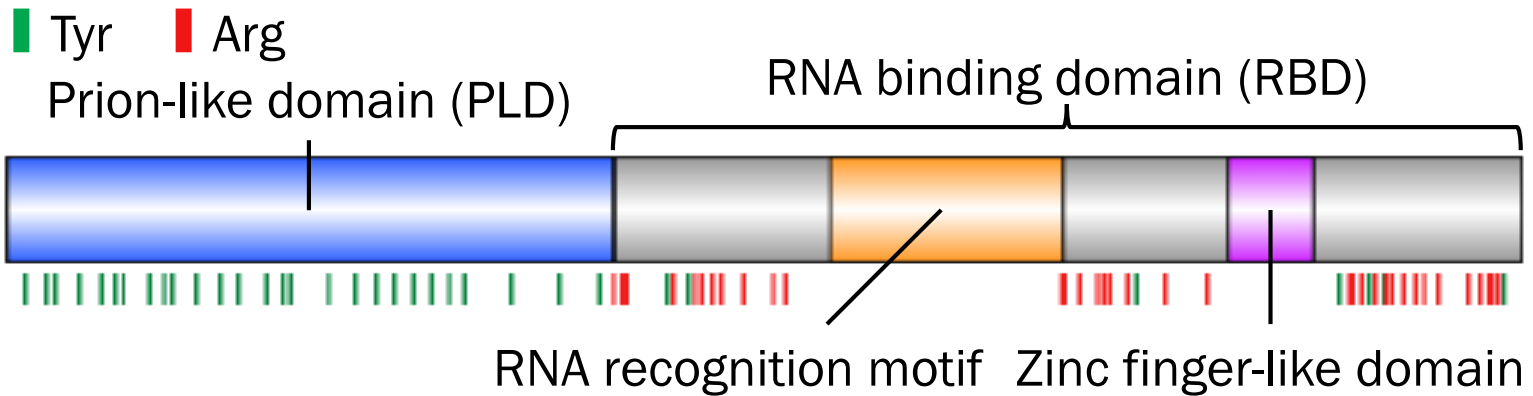
Interplay of Two Domains



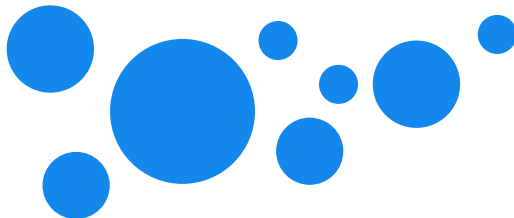
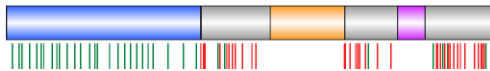
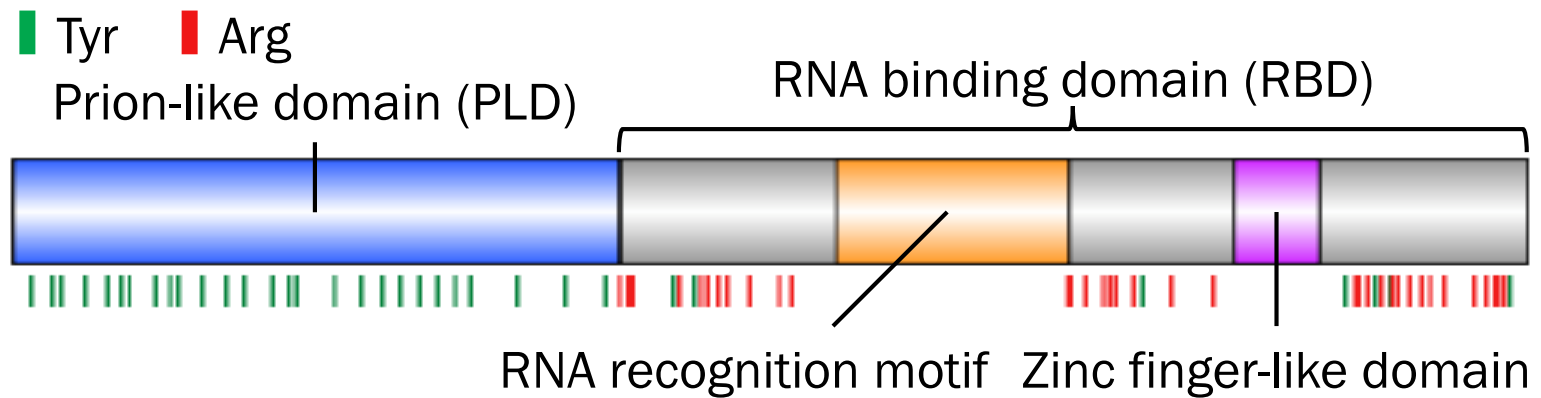
Interplay of Two Domains



Tyrosines and Arginines?



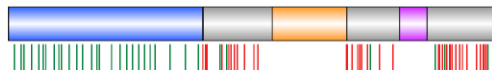
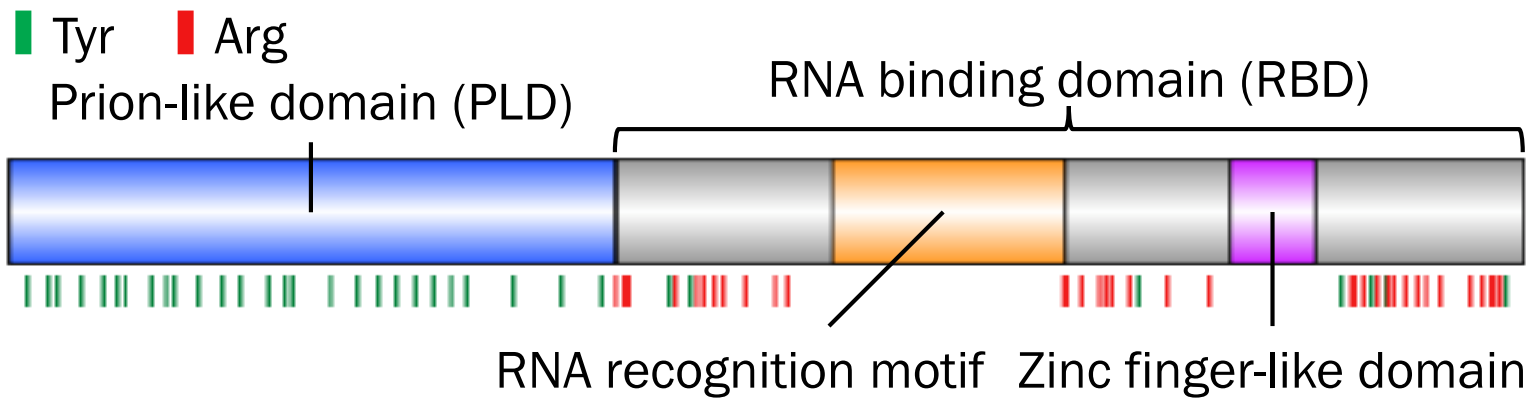
Tyrosines and Arginines?



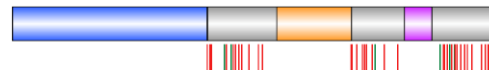
$c_{\text{sat}} = 2 \mu\text{M}$
(75 mM KCl)



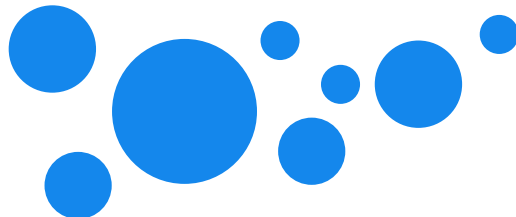
Tyrosines and Arginines?



PLD Tyr → Ser



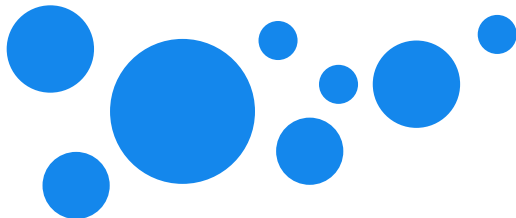
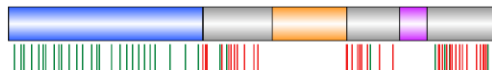
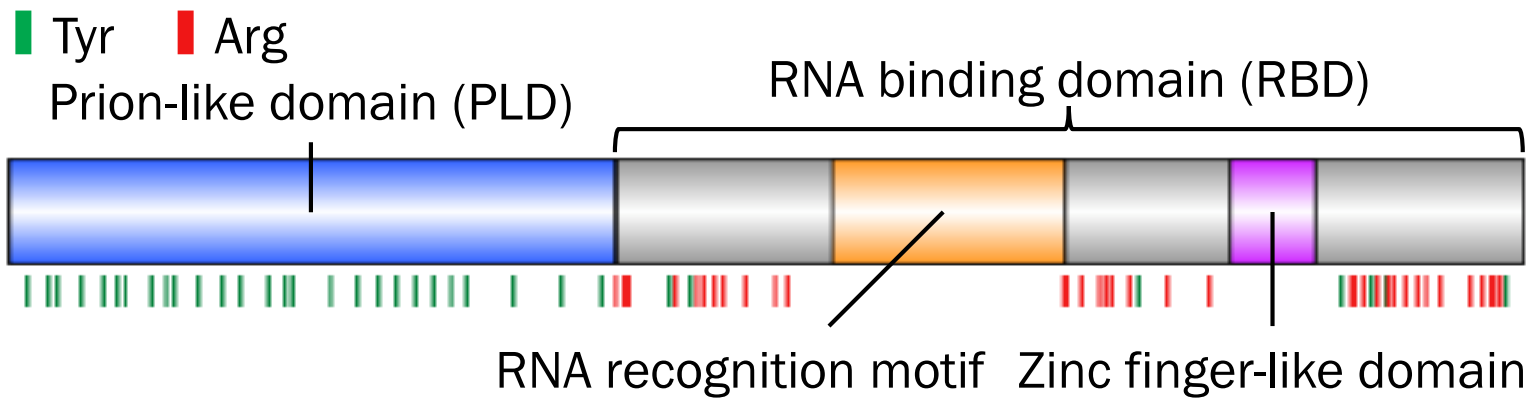
RBD Arg → Gly



$c_{\text{sat}} = 2 \mu\text{M}$
(75 mM KCl)



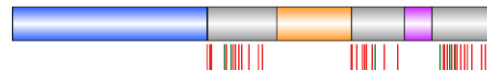
Tyrosines and Arginines?



$C_{\text{sat}} = 2 \mu\text{M}$
(75 mM KCl)



PLD Tyr → Ser



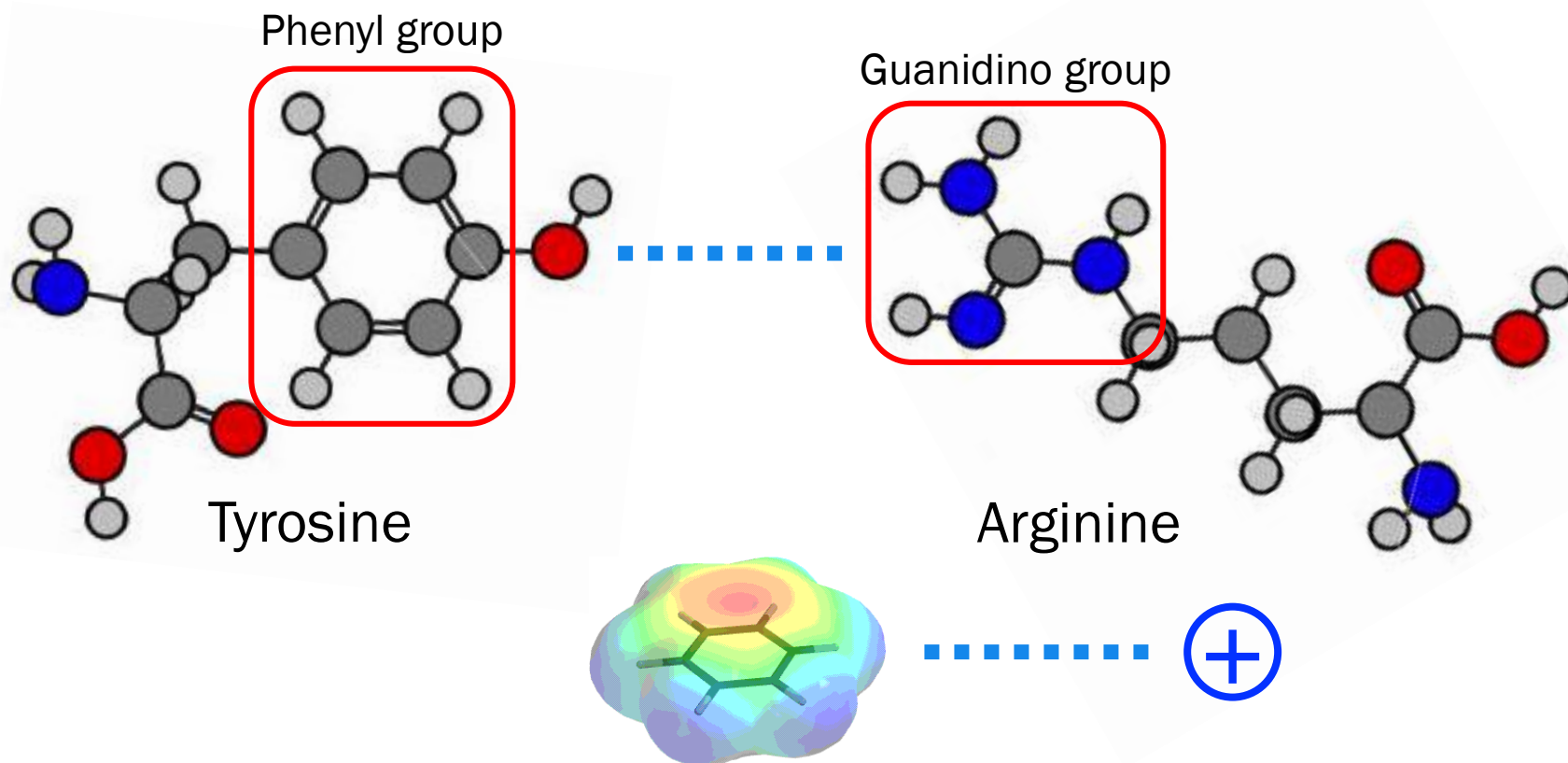
Not detected

RBD Arg → Gly



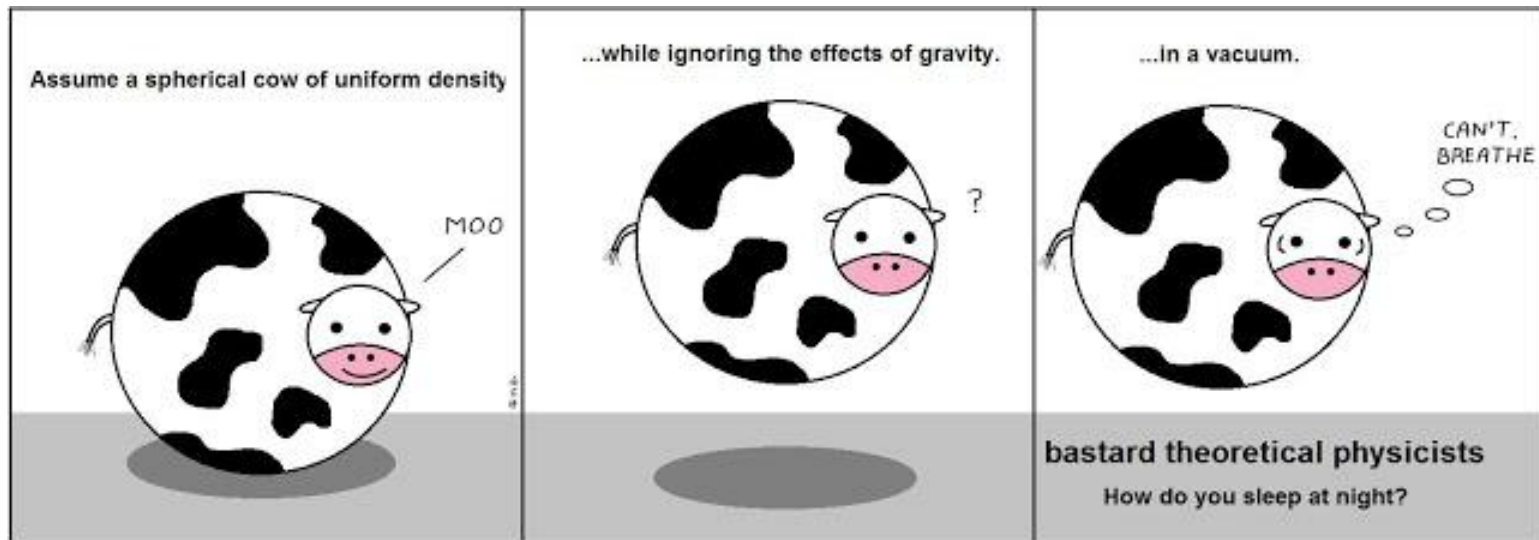
Not detected

Chemistry of Tyr-Arg Interaction



Interaction between π system & positive charge (cation)

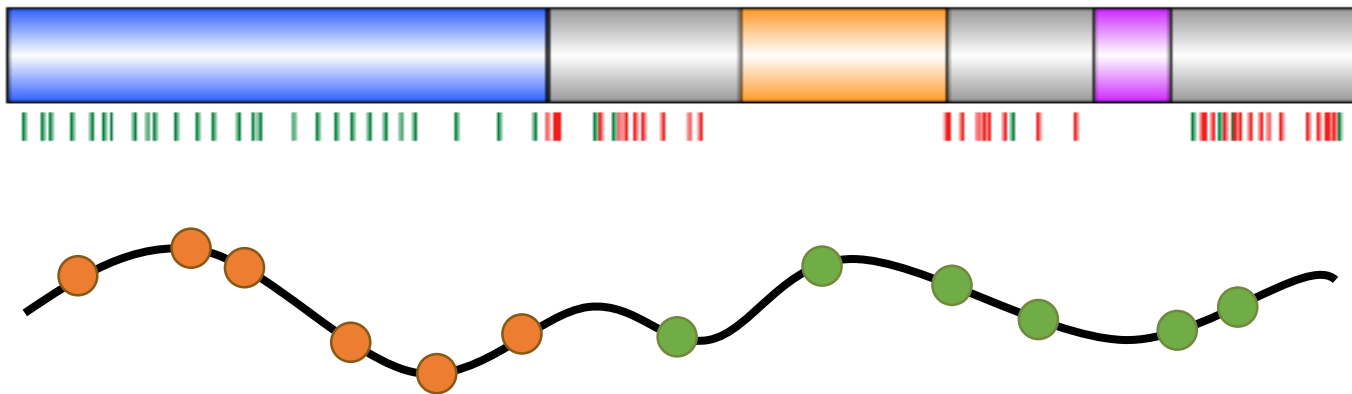
Spherical Cow in Vacuum



Let us start from the **simplest model**

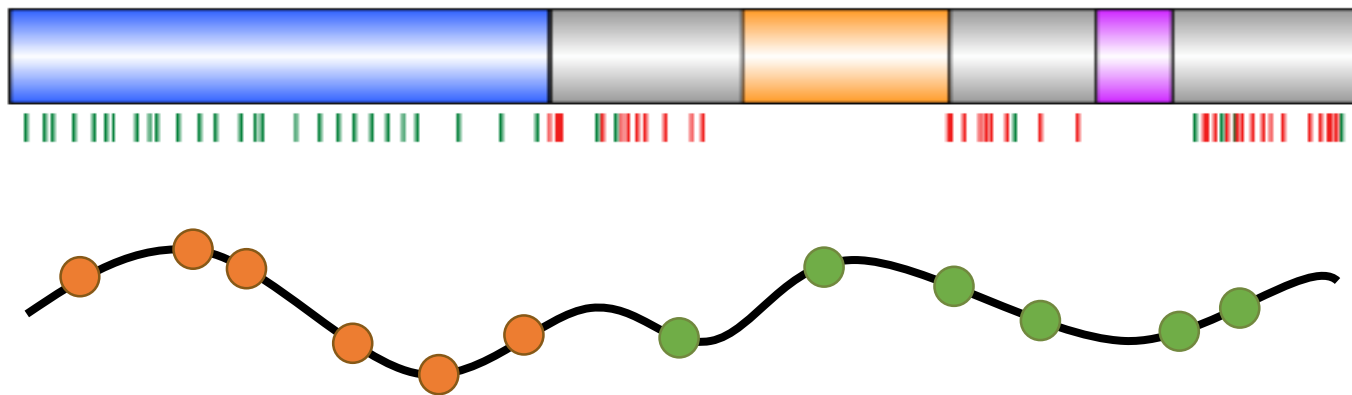
“Stickers and Spacers” Framework

- Phase behaviors come from interplay between
 - **Sticker residues**: chain-chain interactions
 - **Spacer residues**: chain properties

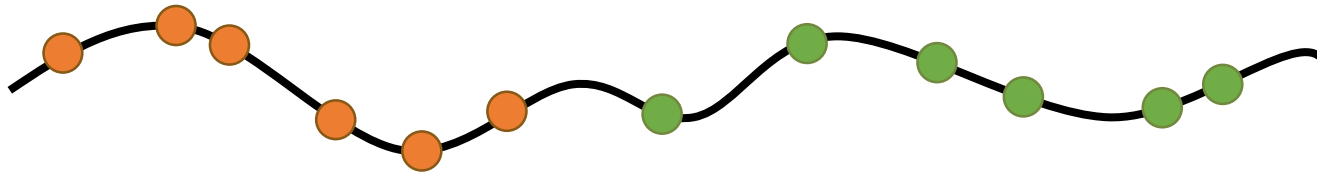


“Stickers and Spacers” Framework

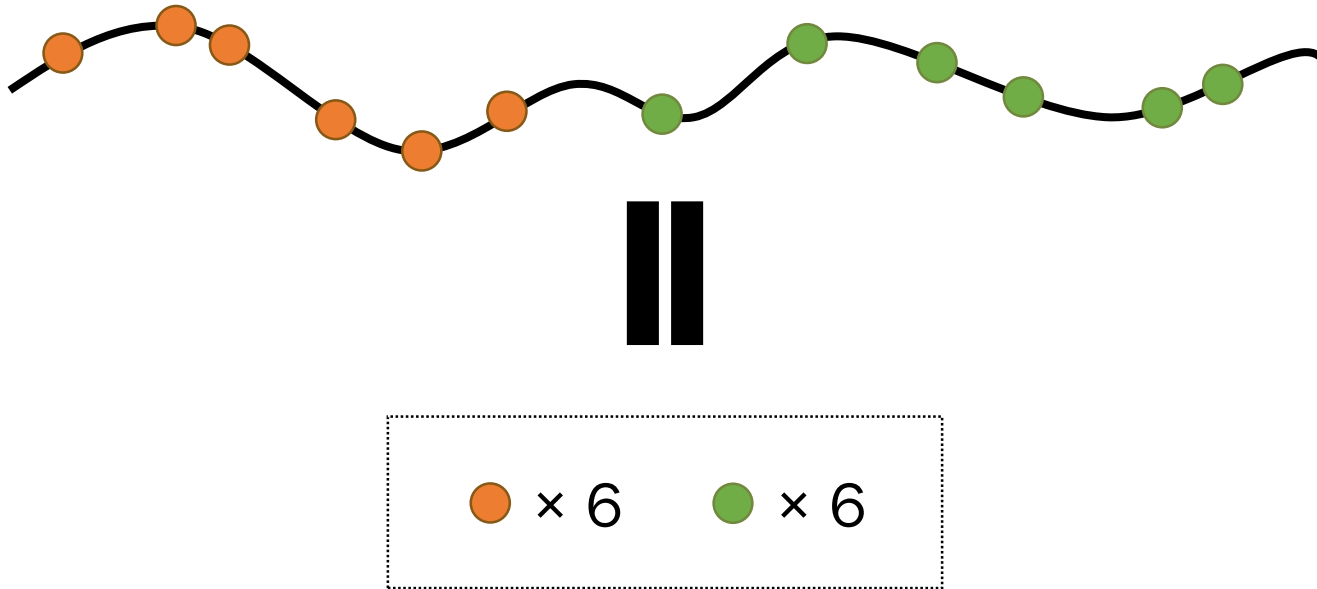
- In the case of FUS
 - Sticker residues: Tyr and Arg
 - Spacer residues: other residues



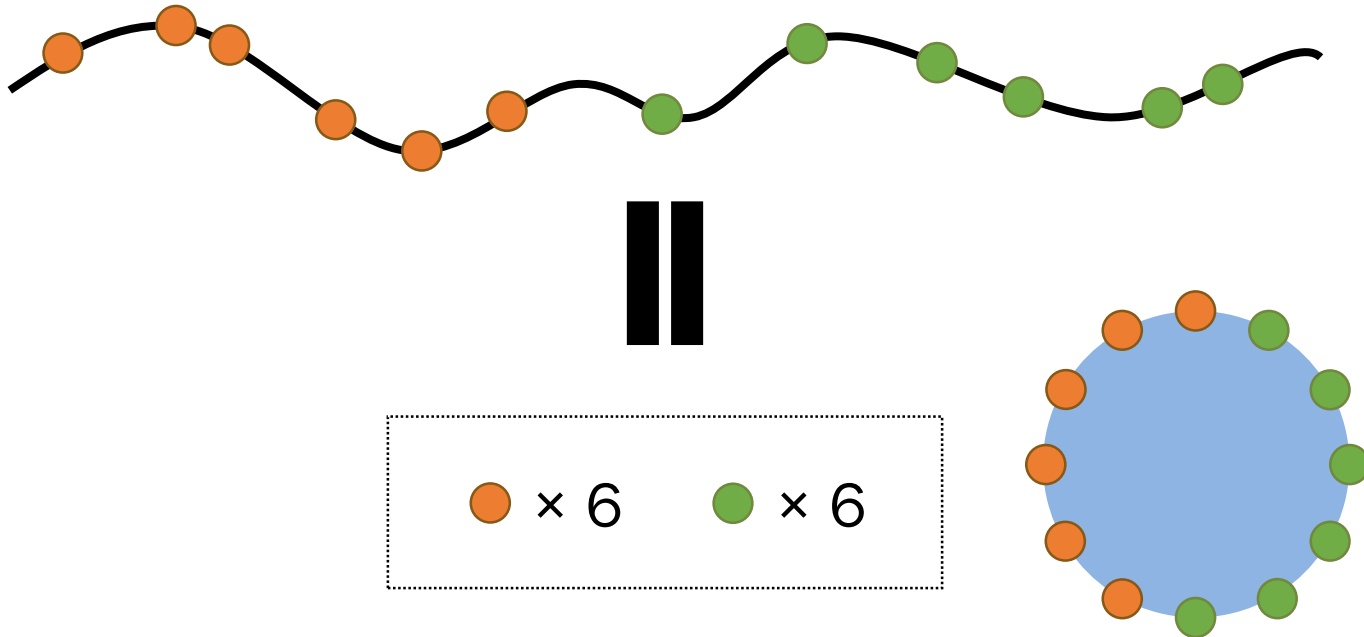
Suppose that the stickers are known ...



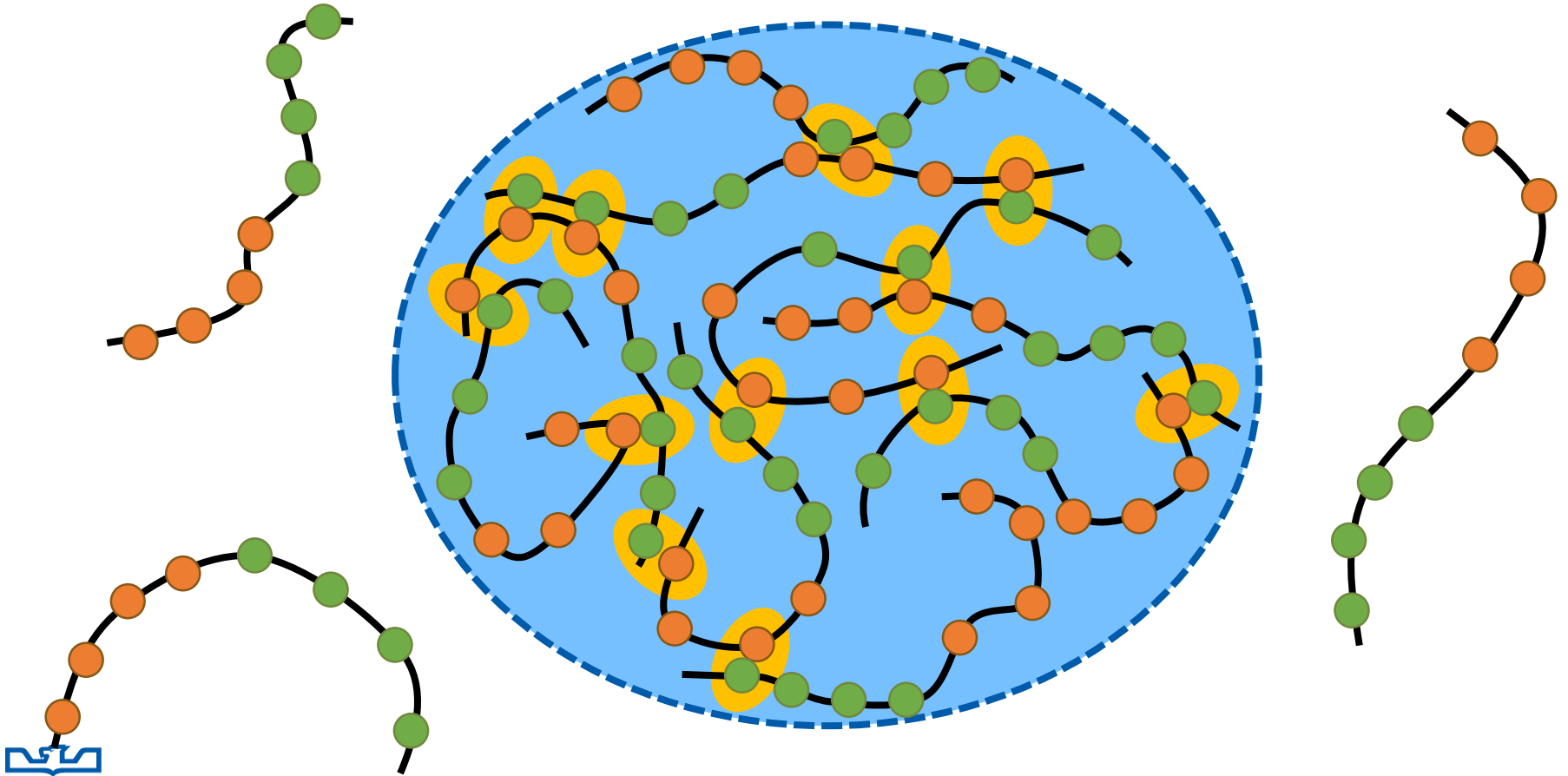
Mean-Field Approach



Mean-Field Approach



Network Formation for Condensation



Mean-Field Prediction

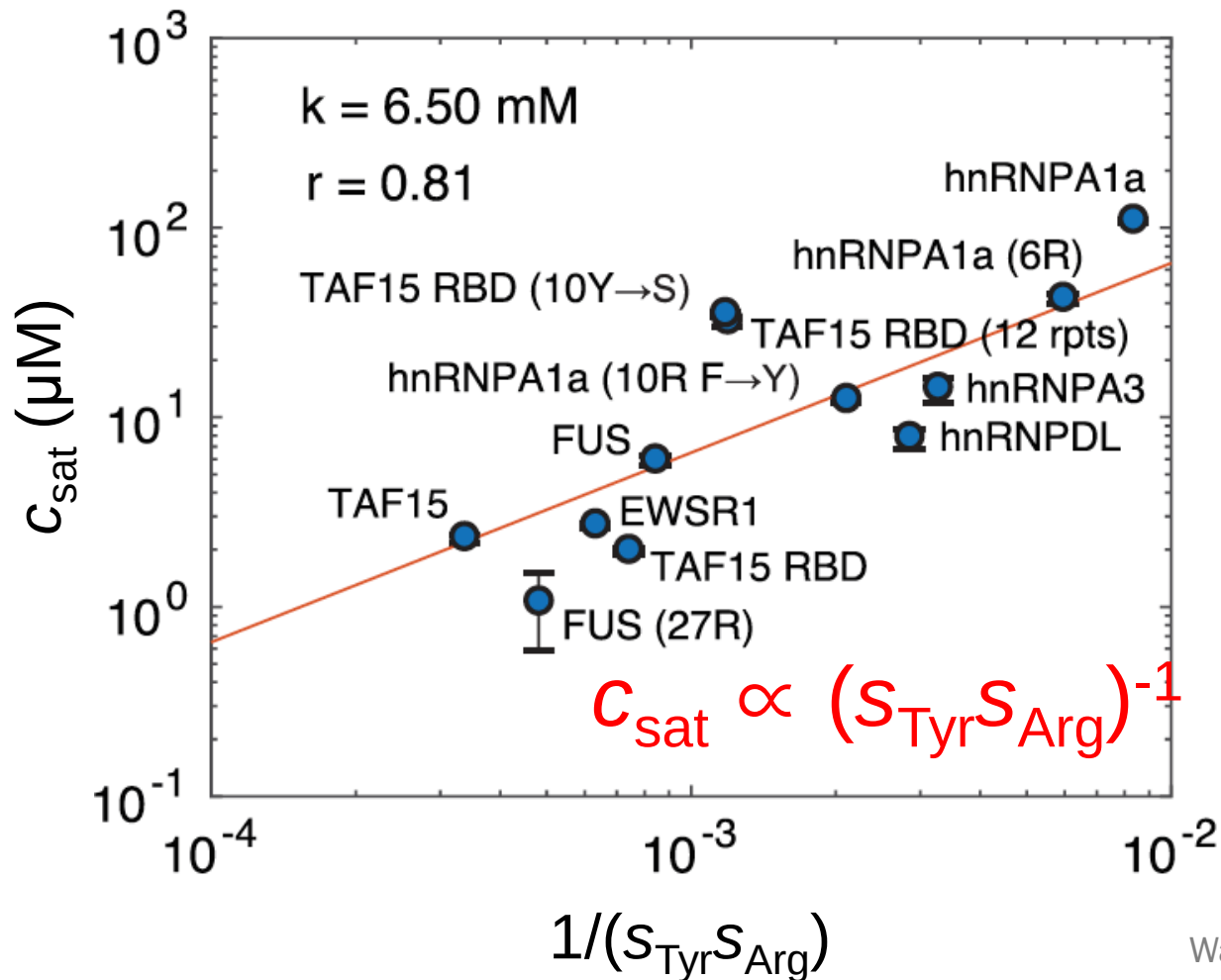


- Two types of stickers (A & B)
- Only A:B interactions allowed

s_A : number of A-stickers & s_B : number of B-stickers

$$C_{\text{sat}} \propto (s_A s_B)^{-1}$$

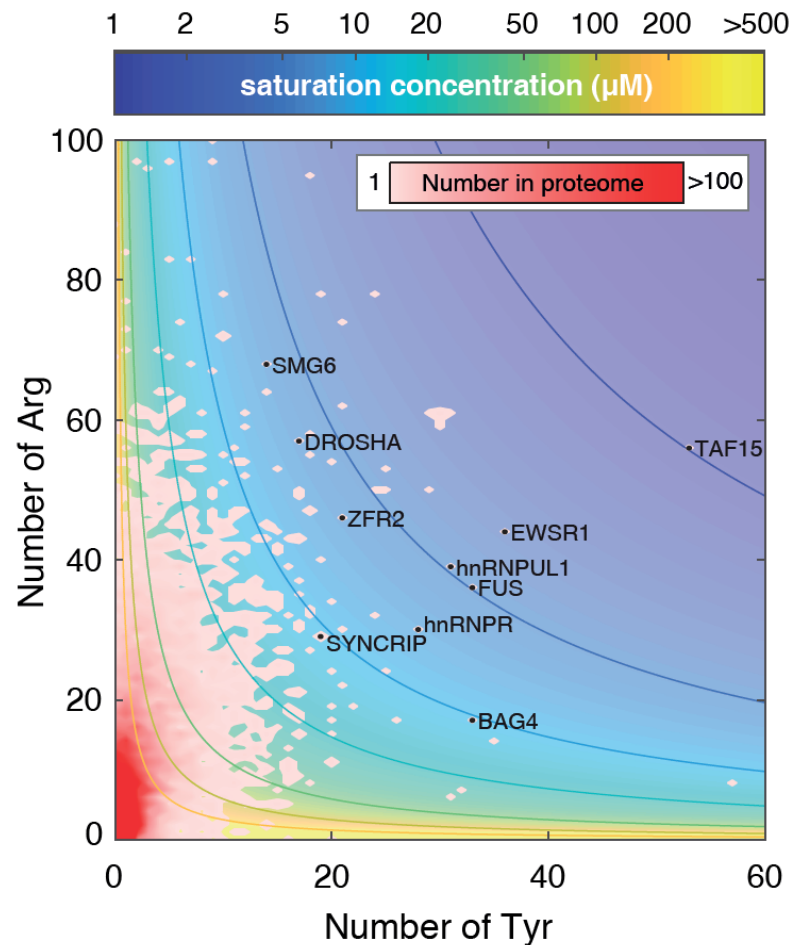
Theory Explains Experiment



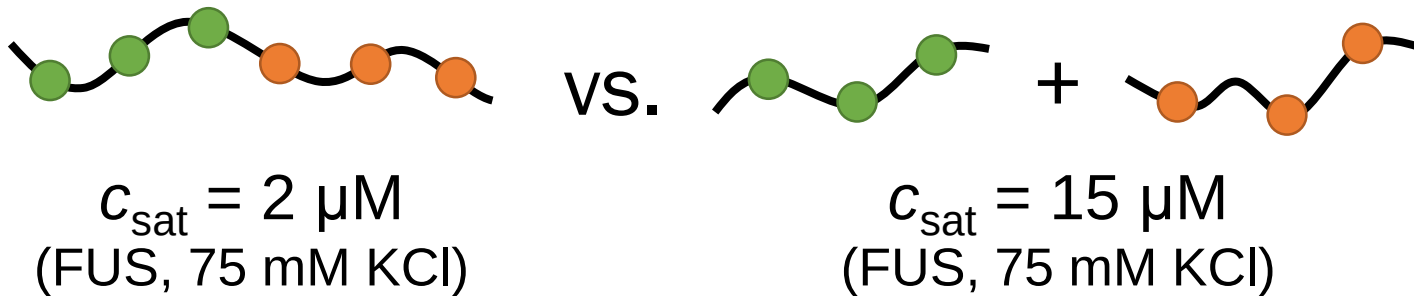
Wang et al., *Cell* (2018)



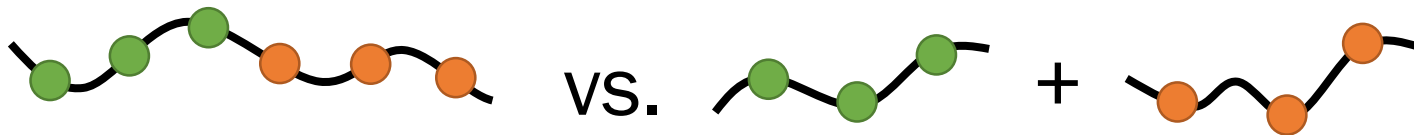
Theory also Gives Predictions



Separation Effect



Qualitative Explanation



$$c_{\text{sat}} = 2 \mu\text{M}$$

(FUS, 75 mM KCl)

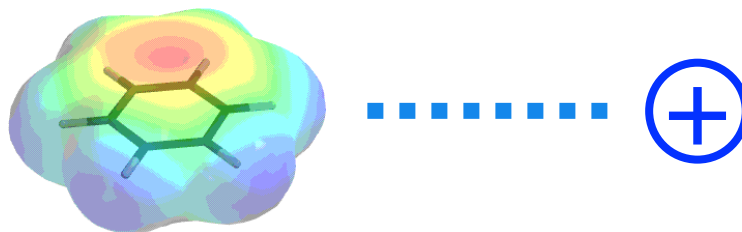
$$c_{\text{sat}} = 15 \mu\text{M}$$

(FUS, 75 mM KCl)

$$c_{\text{sat}} \approx \frac{1}{2\lambda s_A s_B}$$

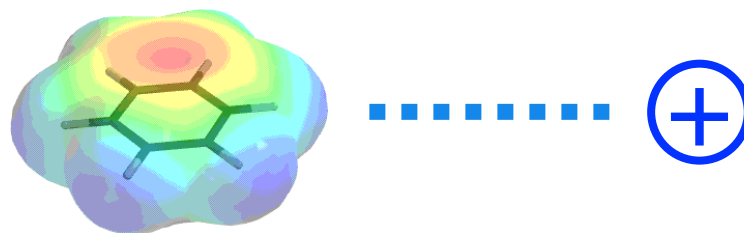
$$c_{\text{sat}} \approx \frac{1}{\lambda s_A s_B}$$

Chemistry of Stickers

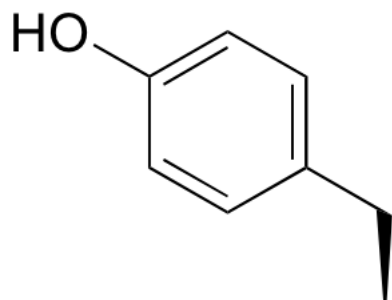


Interaction between π system & positive charge (cation)

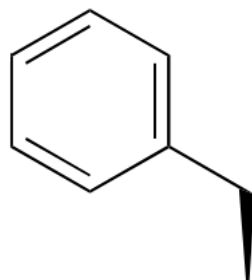
Chemistry of Stickers



Interaction between π system & positive charge (cation)

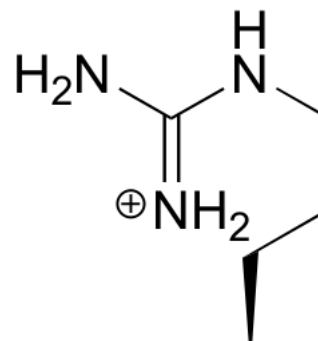


vs.

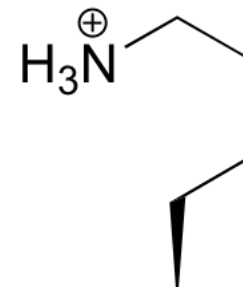


Tyr (Y)

Phe (F)



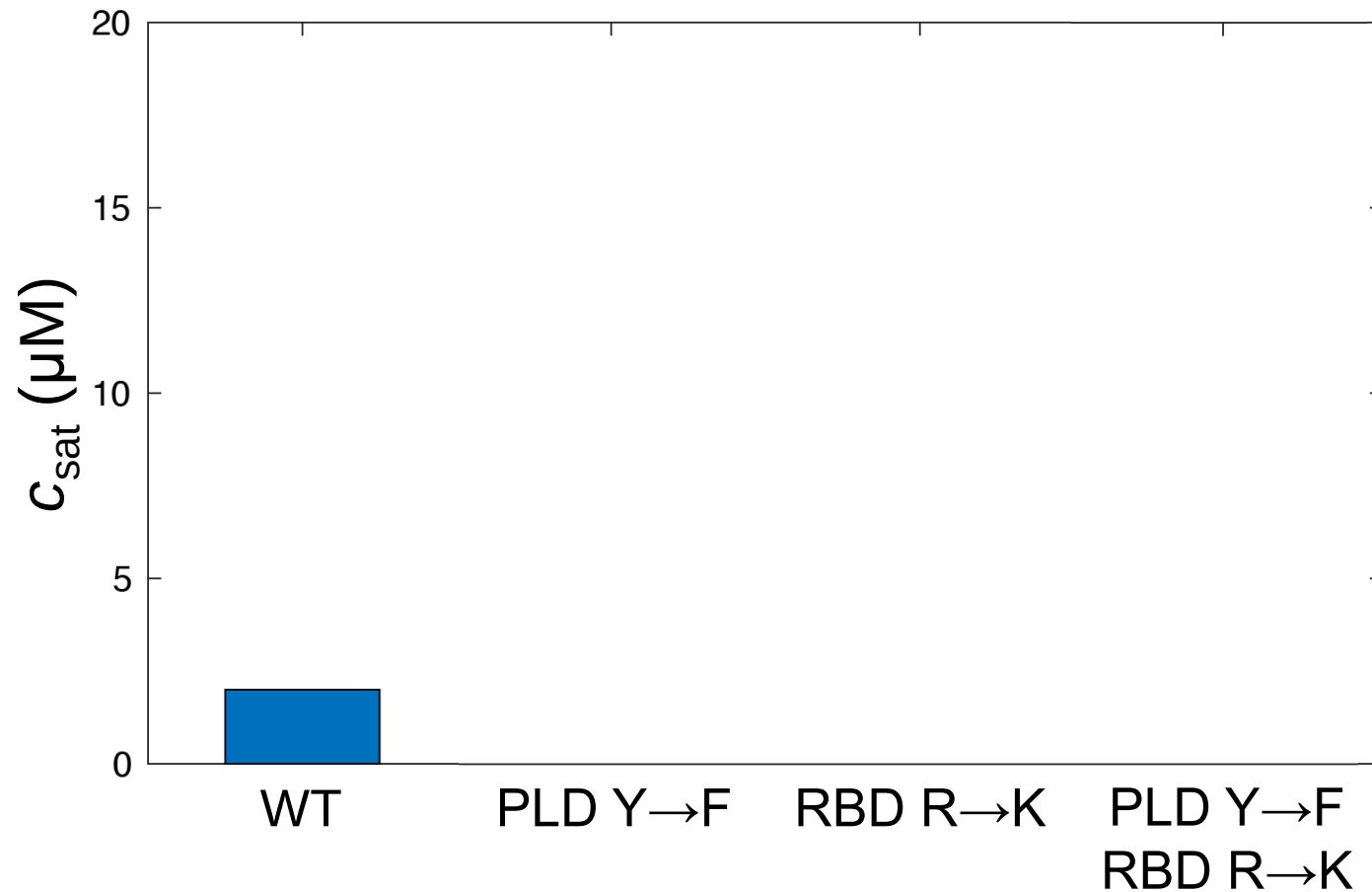
vs.



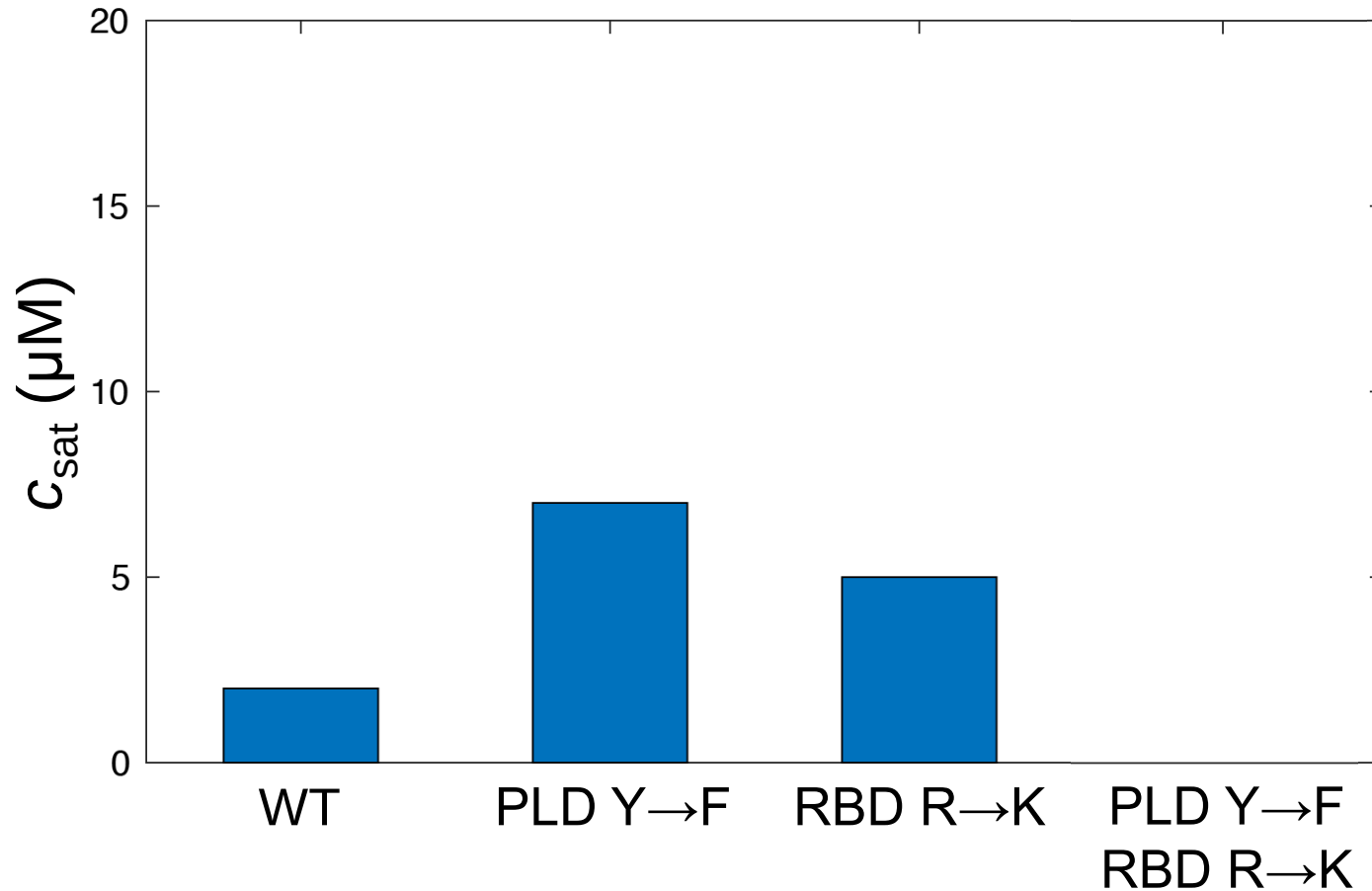
Arg (R)

Lys (K)

FUS Mutants (75 mM KCl)

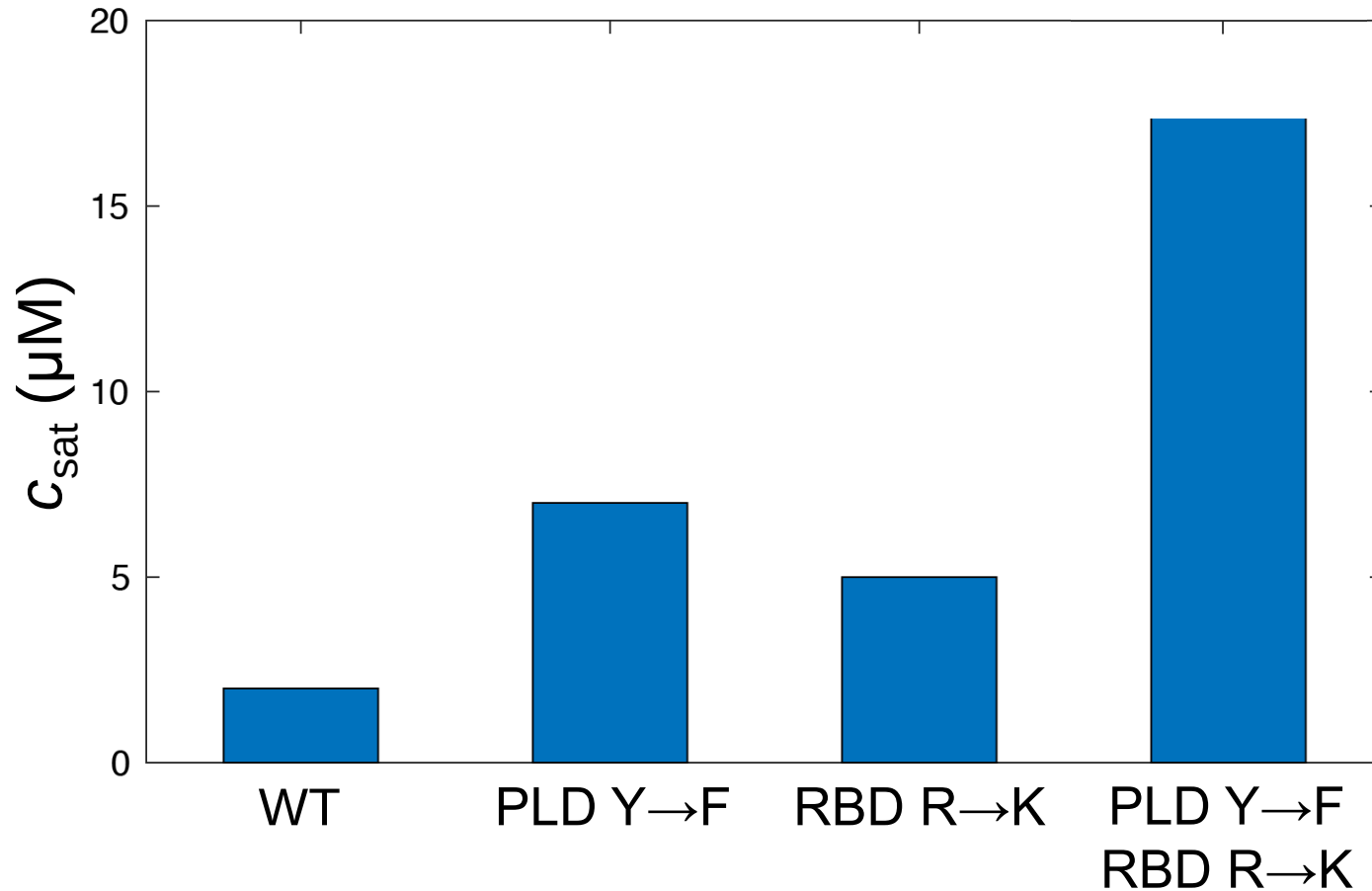


FUS Mutants (75 mM KCl)



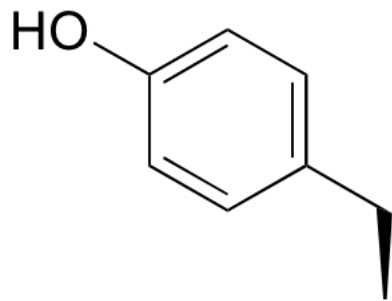
Wang et al., *Cell* (2018)

FUS Mutants (75 mM KCl)



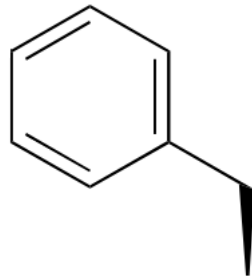
Wang et al., *Cell* (2018)

Sticker Strengths

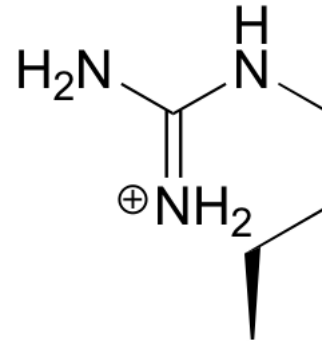


Tyr (Y)

>

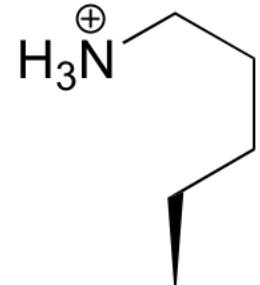


Phe (F)



Arg (R)

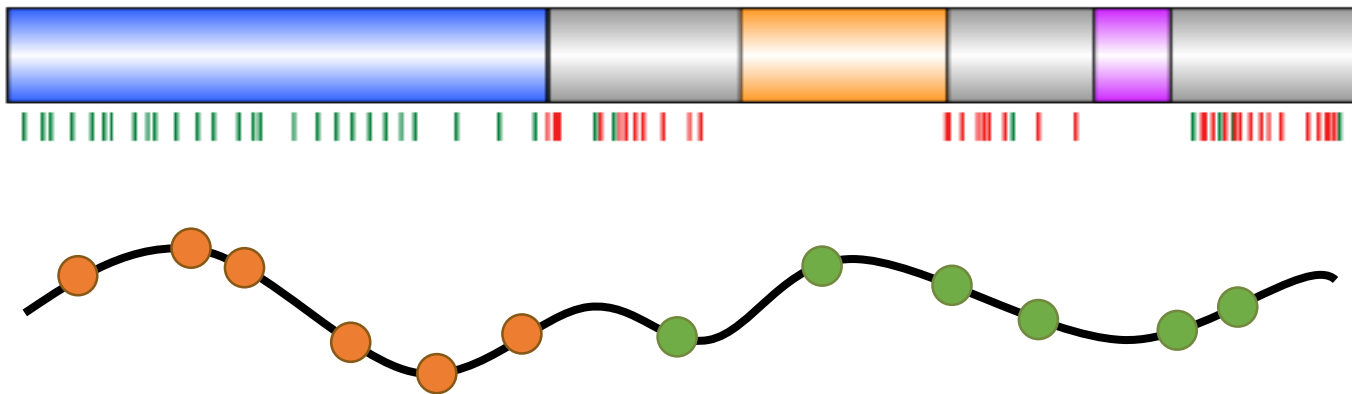
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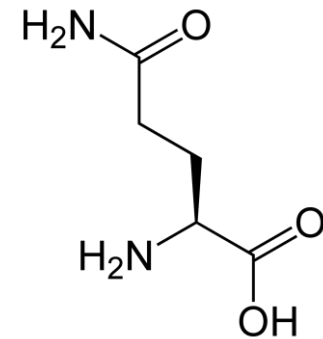
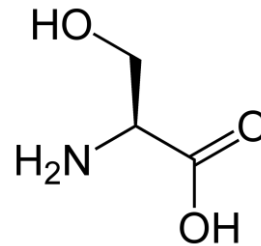
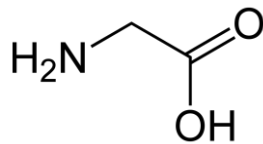
Lys (K)

“Stickers and Spacers” Framework

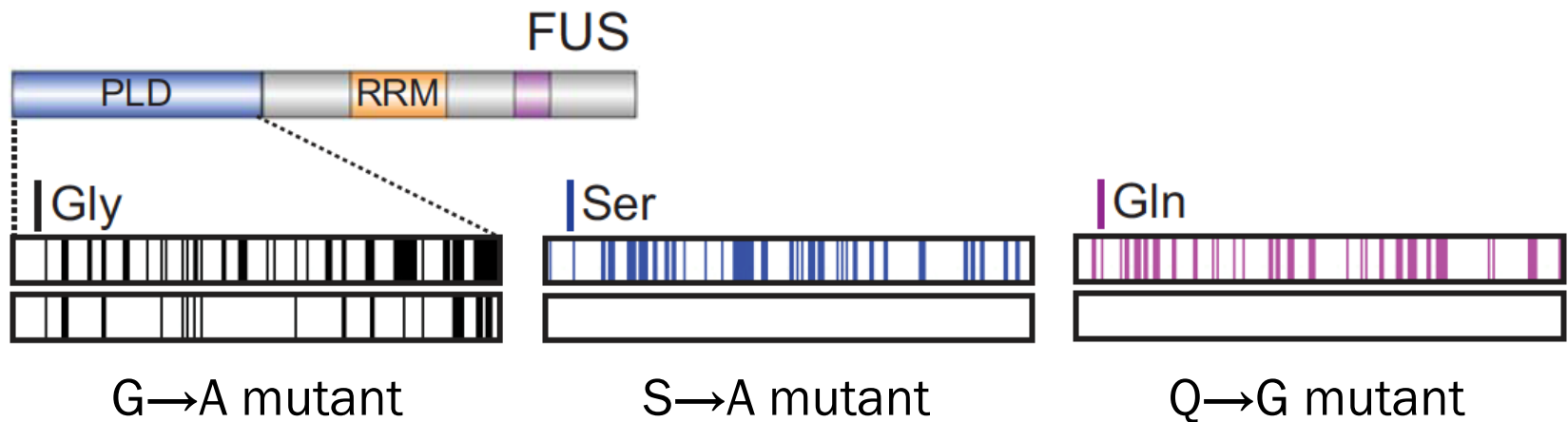
- Phase behaviors come from interplay between
 - **Sticker residues**: chain-chain interactions
 - **Spacer residues**: chain properties



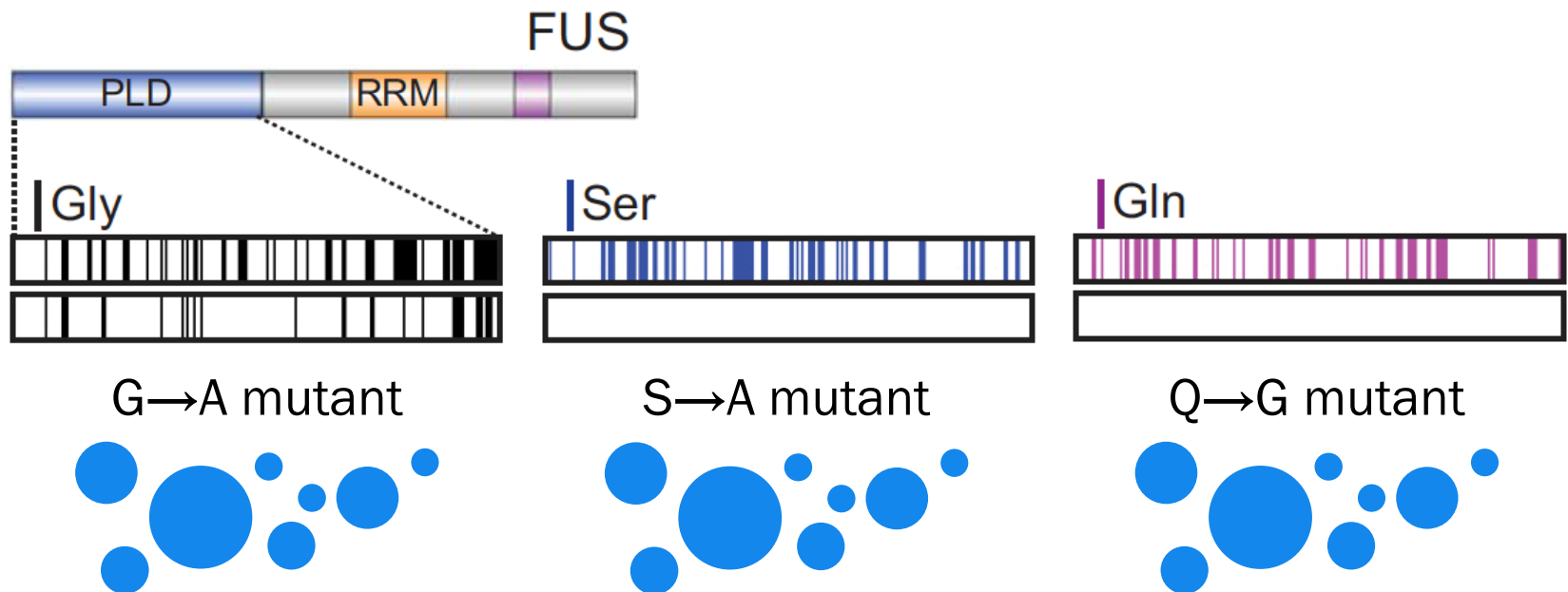
“Spacer” Candidates



Preparation of Mutants



They All do not Change c_{sat}



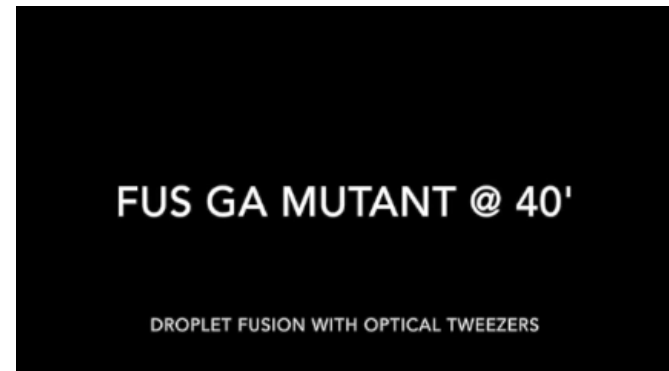
$c_{\text{sat}} = 1-2 \mu\text{M}$ (at 75 mM KCl)

What do Spacers do? Case of G

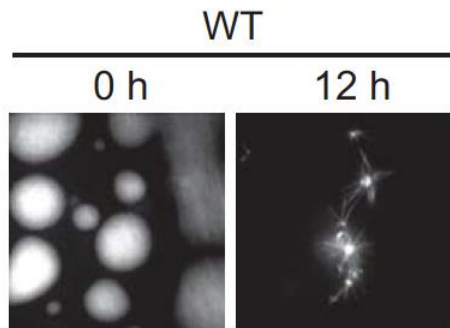
Wildtype @ 120 min



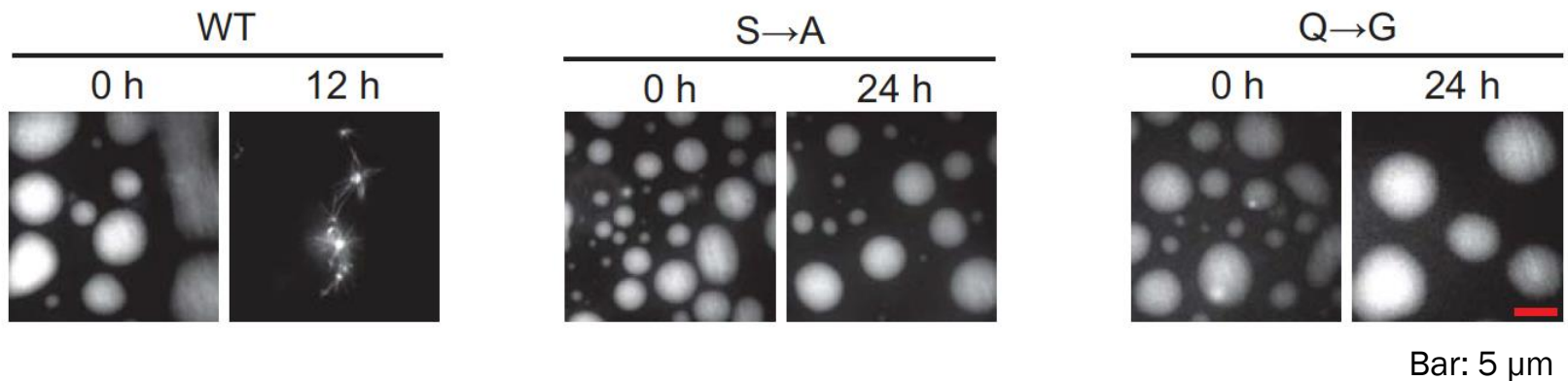
G→A mutant @ 40 min



What do Spacers do? Case of S and Q



What do Spacers do? Case of S and Q



“Stickers and Spacers” Framework

- Phase behaviors come from interplay between
 - **Sticker residues**: chain-chain interactions
 - **Spacer residues**: chain properties

